Equilibrium Bifurcations and Periodic Dynamics in a Planar Restricted Five-Body Problem with Four Unequal Primaries

Muhammad Shoaib* A. R. Kashif†

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Abstract

We investigate the planar restricted five-body problem in which four unequal primaries form a non-symmetric coplanar central configuration and an infinitesimal fifth body moves in the associated uniformly rotating field. After implementing and validating the model, we fix $(a, b) = (0.9, -0.1)$ and continue the admissible central configurations in the remaining geometric parameters. The principal result is structural: on this slice the admissible set $\psi = 0$ separates numerically into two distinct connected components, so configurations with different equilibrium multiplicities need not belong to a single continuous family. Pseudo-arclength continuation combined with independent multi-start equilibrium searches reveals the interior cascades

$$5 \to 7 \to 5 \qquad \text{and} \qquad 7 \to 9 \to 11 \to 9 \to 7 \to 5.$$

All seven interior count changes are generic saddle-node folds, verified through an augmented degeneracy system, nondegeneracy conditions, and the local scaling

$$d \propto |\ell - \ell_c|^{1/2}.$$

The eleven-equilibrium regime exhibits a branch exchange: the pair created at the $9 \to 11$ fold is distinct from the pre-existing pair destroyed at the subsequent $11 \to 9$ fold. Both admissible components contain reliably resolved intervals of spectrally stable equilibria. From selected simple centre modes we continue ten Lyapunov periodic-orbit families comprising 198 corrected periodic solutions with closure residuals below 10^{-10}. Families continued from the selected doubly-elliptic parents remain Floquet stable over the computed intervals, whereas those continued from selected saddle×centre parents are real unstable. Selected near-commensurate families approach the period-doubling threshold; the closest, on family 10, reaches $|\nu+2| \simeq 10^{-7}$ but turns back without crossing. Finally, critical Jacobi levels are used to demonstrate a resolution-robust representative change in Hill-region connectivity and to place selected periodic families within their accessible regions. All conclusions concerning central-configuration connectivity and equilibrium multiplicity are numerical and specific to the investigated parameter slice.

1 Introduction

Restricted n-body models, in which an infinitesimal body moves in the gravitational field of a prescribed relative equilibrium of massive primaries, provide a classical setting for studying the organization of celestial-mechanical phase space. In the circular restricted three-body problem (CR3BP), the five Lagrange equilibria, their linear stability, the associated zero-velocity curves

*Affiliation to be finalized.
†Affiliation to be finalized.

and Hill regions, and the periodic-orbit families emanating from them are well understood [12, 6]. Higher restricted problems retain this basic structure but introduce an additional dependence on the geometry and mass distribution of the primaries: both the number of relative equilibria and their dynamical character can change as the primary configuration is varied.

This dependence is already apparent in the restricted four-body problem. For three primaries in the Lagrange equilateral configuration, the locations, number, and stability of the libration points vary with the primary mass ratios, and the problem supports extensive families of periodic orbits [1, 2]; the stability of the corresponding four-body relative equilibria was analysed earlier by Simó [10]. Baltagiannis and Papadakis also extended the periodic-orbit analysis to configurations with three unequal primaries, including the linear stability of the resulting families [2]. In contrast, we are not aware of an analogous periodic-orbit study in the restricted five-body problem in which all four primaries have distinct masses and form a general non-symmetric central configuration.

Restricted five-body models introduce a further level of geometric freedom. Ollongren considered four primaries in a symmetric configuration and obtained nine libration points, including stable equilibria [8]. Subsequent studies have examined regular-polygon and other symmetric arrangements [5], numerical basins of attraction of the equilibria [14], and networks of symmetric and non-symmetric periodic orbits [13]. These investigations demonstrate that the restricted five-body problem can possess a substantially richer equilibrium and periodic-orbit structure than the CR3BP.

A crucial additional constraint is that the primaries cannot, in general, be placed arbitrarily. To remain fixed relative to one another in a uniformly rotating frame, they must themselves form a central configuration (CC). The geometry and multiplicity of four-body central configurations have therefore long been studied independently of the restricted problem, from the classical work of Moulton [7] to later results on their stability, bifurcation structure, and finiteness [10, 4]. Many tractable four-body families impose some form of symmetry, such as equal masses, reflection symmetry, or a kite or trapezoidal geometry [11, 3]. Such assumptions reduce the dimensionality of the CC problem and can permit partly analytic treatment, but they also restrict the set of primary geometries accessible to the infinitesimal body.

The general coplanar four-body formulation of Shoaib et al. [9] removes this restriction. After normalization, three primaries define a reference triangle while the fourth is free in the plane; the four masses are then determined, up to scale, by a scalar central-configuration constraint together with positivity conditions on the resulting mass ratios. The admissible primary configurations therefore form a nontrivial subset of a continuous geometric parameter space rather than a prescribed symmetric family. Introducing an infinitesimal fifth body into this rotating four-primary configuration leads naturally to the question considered here:

> *How do the equilibrium structure and the associated local nonlinear dynamics of the restricted body change as the underlying non-symmetric four-primary central configuration is continuously deformed?*

This question cannot be answered by comparing isolated parameter choices alone. Two configurations may possess different numbers of equilibria without belonging to the same connected component of the admissible CC set, in which case there is no bifurcation sequence directly connecting their equilibrium counts. Conversely, an individual admissible component may pass through equilibrium multiplicities not represented by the original sampled configurations. The topology of the primary CC set must therefore be determined together with the restricted equilibria if one wishes to understand how those equilibria are created, destroyed, and reorganized.

To our knowledge, existing studies of unequal or non-symmetric restricted four- and five-body problems predominantly analyse specified primary configurations or restricted parameter families rather than tracing the equilibrium structure along a continuously varying general four-body CC. Detailed continuation and periodic-orbit studies of restricted five-body models have, in particular, largely exploited prescribed symmetric primary configurations [8, 5, 13]. We are not aware of a study that simultaneously follows an admissible non-symmetric four-primary CC family, determines the restricted equilibria along it, locates and classifies their bifurcations, tracks their linear stability, and then continues the periodic dynamics associated with the resulting centre modes. This is the gap addressed in the present work.

Our analysis combines pseudo-arclength continuation of the primary CC constraint with an independent global search for the equilibria of the infinitesimal body at each continuation point. The distinction is important: an equilibrium branch created at a saddle-node does not exist before the bifurcation and may therefore be missed if one merely continues previously known roots. Changes in the equilibrium count are bracketed and the corresponding critical configurations are refined by solving the equilibrium equations together with the Hessian-degeneracy condition. Genericity is tested through the fold transversality and quadratic-nondegeneracy conditions and independently through the local square-root scaling of the merging pair. Linear spectra are then followed along the resulting equilibrium branches. Centre modes at selected equilibria provide initial seeds for differential correction and continuation of nonlinear periodic-orbit families, whose stability is determined from their Floquet multipliers. Finally, the critical Jacobi values provide a geometric link between the equilibrium structure and the accessible Hill regions. The principal results are as follows.

1. On the common Case-1/Case-2 slice
$$(a, b) = (0.9, -0.1),$$
the admissible four-primary CC set separates numerically into two distinct connected components. Consequently, the five- and nine-equilibrium reference configurations do not lie on a single fixed-(a, b) continuation family.

2. Equilibrium multiplicity changes along the two admissible components through seven generic saddle-node folds. The corresponding sequences are
$$5 \to 7 \to 5$$
on the first component and
$$7 \to 9 \to 11 \to 9 \to 7 \to 5$$
on the second. The eleven-equilibrium regime contains a nontrivial branch exchange: the pair created at the $9 \to 11$ fold is distinct from the pre-existing pair destroyed at the subsequent $11 \to 9$ fold.

3. Both admissible components contain intervals of spectrally stable equilibria. Continuation from selected centre modes yields ten Lyapunov periodic-orbit families. The families associated with the selected doubly-elliptic parent equilibria remain Floquet stable over the computed intervals, whereas those associated with selected saddle$\times$centre parents are real unstable. Families seeded near low-order centre-frequency commensurabilities approach the period-doubling threshold but do not cross it within the computed continuation range.

4. The critical Jacobi values associated with the equilibrium branches provide a direct connection between the bifurcation structure and the geometry of the accessible region. A representative Hill-region connectivity change is resolved numerically and related to the computed periodic motion.

The principal conclusion is therefore not simply that the present asymmetric problem supports as many as eleven equilibria. Rather, on the investigated parameter slice, the *topology of the admissible primary central-configuration set organizes which restricted equilibrium branches coexist and which bifurcation sequences are accessible.* The eleven-equilibrium state, the stable equilibrium intervals, and the associated periodic-orbit families arise as dynamical consequences of this organization. All statements concerning component connectivity and equilibrium completeness are numerical and apply to the chosen slice of the full four-parameter admissible set; continuation through the complete CC parameter space remains an open problem.

2 Mathematical model

We describe the model in normalized form. The four-body central-configuration formulation of Section 2.1 is due to [9] and is *not* claimed as a result of the present paper; we restate it only to fix notation and because it defines the parameter family along which the restricted problem is posed. The restricted fifth-body model follows in Sections 2.2–2.3.

2.1 The primary central configuration

Following the normalization of Shoaib et al. [9], we use the translation, rotation, reflection, and scaling invariances of the planar central-configuration problem to place the four primaries at

$$\mathbf{r}_0 = (0, b), \qquad \mathbf{r}_1 = (-1, 0), \qquad \mathbf{r}_2 = (0, a), \qquad \mathbf{r}_3 = (c_1, c_2), \tag{1}$$

with

$$a > 0, \qquad b < a, \qquad b \neq \pm a.$$

The choice $\mathbf{r}_1 = (-1, 0)$ fixes the length scale, while $\mathbf{r}_0$ and $\mathbf{r}_2$ lie on the y-axis and $\mathbf{r}_3 = (c_1, c_2)$ remains free in the plane. The geometry is therefore described by the parameter vector

$$\mathbf{p} = (a, b, c_1, c_2).$$

For compactness, define

$$\begin{aligned} &\sigma = (1 + a^2)^{3/2}, && \varsigma = (1 + b^2)^{3/2}, && \tau = (a - b)^3, \\ &G = \left[c_1^2 + (c_2 - a)^2\right]^{3/2}, && T = \left[c_1^2 + (c_2 - b)^2\right]^{3/2}, && M = \left[(c_1 + 1)^2 + c_2^2\right]^{3/2}. \end{aligned} \tag{2}$$

These quantities are the cubes of the six mutual distances between the primaries.

Let

$$\rho_0 = \frac{m_0}{m_3}, \qquad \rho_1 = \frac{m_1}{m_3}, \qquad \rho_2 = \frac{m_2}{m_3}$$

denote the primary mass ratios. For the nondegenerate geometries considered here, the coplanar four-body central-configuration condition of [9] reduces to the scalar equation

$$\begin{aligned} \psi(a, b, c_1, c_2) = {} & \tau(\sigma - \varsigma)M + T\left[(\tau - \sigma)M + \sigma(\varsigma - \tau)\right] \\ & + G\left[(\sigma - \varsigma)T + (\varsigma - \tau)M + \varsigma(\tau - \sigma)\right] = 0, \end{aligned} \tag{3}$$

and the corresponding mass ratios are

$$\rho_0 = \frac{(c_2 - a - ac_1)(M - G)\varsigma\tau}{(a - b)(\varsigma - \tau)MG}, \qquad \rho_1 = \frac{c_1(G - T)\sigma\varsigma}{(\sigma - \varsigma)GT}, \qquad \rho_2 = \frac{(c_2 - b - bc_1)(M - T)\sigma\tau}{(a - b)(\tau - \sigma)MT}. \tag{4}$$

Equations (3)–(4) are the four-body central-configuration result of [9]; they are restated here only to define the family of primary configurations used in the restricted problem.

Not every solution of $\psi = 0$ corresponds to a positive-mass, nondegenerate configuration. We therefore define the admissible set

$$\mathcal{A} = \{\mathbf{p} : \psi(\mathbf{p}) = 0,\ \rho_0(\mathbf{p}) > 0,\ \rho_1(\mathbf{p}) > 0,\ \rho_2(\mathbf{p}) > 0\}, \tag{5}$$

subject additionally to non-vanishing denominators in (4) and the exclusion of primary collisions. Since only mass ratios enter the central-configuration construction, the overall mass scale remains arbitrary. We fix this scale by setting $m_3 = 1$, so that the normalized primary masses used throughout the restricted problem are

$$(m_0, m_1, m_2, m_3) = (\rho_0, \rho_1, \rho_2, 1).$$

2.2 The restricted fifth body: potential, equilibria, and linear stability

For each admissible primary configuration $\mathbf{p} \in \mathcal{A}$, the four primary masses and positions are fixed in the uniformly rotating frame, and the fifth body is taken to be infinitesimal: it moves in their gravitational field but does not perturb the primary central configuration. Let (x, y) denote its rotating-frame coordinates, and define the distances to the four primaries by

$$\begin{aligned} r_{40} &= \sqrt{x^2 + (y-b)^2}, \quad r_{41} = \sqrt{(x+1)^2 + y^2}, \\ r_{42} &= \sqrt{x^2 + (y-a)^2}, \quad r_{43} = \sqrt{(x-c_1)^2 + (y-c_2)^2}. \end{aligned} \tag{6}$$

With the rotation rate normalized to unity, the effective potential is

$$\Omega(x, y) = \frac{1}{2}(x^2 + y^2) + \sum_{i=0}^{3} \frac{m_i}{r_{4i}}, \tag{7}$$

where

$$(m_0, m_1, m_2, m_3) = (\rho_0, \rho_1, \rho_2, 1).$$

The equations of motion are

$$\ddot{x} - 2\dot{y} = \Omega_x, \qquad \ddot{y} + 2\dot{x} = \Omega_y, \tag{8}$$

with

$$\Omega_x = x - \sum_{i=0}^{3} m_i \frac{x - x_i}{r_{4i}^3}, \qquad \Omega_y = y - \sum_{i=0}^{3} m_i \frac{y - y_i}{r_{4i}^3}, \tag{9}$$

where (x_i, y_i) are the primary positions defined in (1).

The relative equilibria, or libration points, of the infinitesimal body are the stationary solutions of the rotating-frame equations,

$$\Omega_x(x, y) = 0, \qquad \Omega_y(x, y) = 0. \tag{10}$$

Unlike the five Lagrange points of the CR3BP, the number and locations of these equilibria depend on the primary configuration $\mathbf{p} \in \mathcal{A}$. Their variation along the admissible central-configuration set is the equilibrium bifurcation problem studied in Section 4.

To analyse both the local root structure of (10) and the linear dynamics near an equilibrium, we use the Hessian of Ω. Writing

$$\Delta x_i = x - x_i, \qquad \Delta y_i = y - y_i,$$

its entries are

$$
\begin{aligned}
\Omega_{xx} &= 1 - \sum_{i=0}^{3} m_i \left(\frac{1}{r_{4i}^3} - \frac{3\Delta x_i^2}{r_{4i}^5} \right), \\
\Omega_{yy} &= 1 - \sum_{i=0}^{3} m_i \left(\frac{1}{r_{4i}^3} - \frac{3\Delta y_i^2}{r_{4i}^5} \right), \\
\Omega_{xy} &= 3 \sum_{i=0}^{3} m_i \frac{\Delta x_i \Delta y_i}{r_{4i}^5}.
\end{aligned}
\tag{11}
$$

Thus

$$
H_\Omega = \begin{pmatrix} \Omega_{xx} & \Omega_{xy} \\ \Omega_{xy} & \Omega_{yy} \end{pmatrix}
\tag{12}
$$

is simultaneously the Hessian of the effective potential and the Jacobian of the equilibrium map

$$
F(x, y) = \begin{pmatrix} \Omega_x \\ \Omega_y \end{pmatrix}.
$$

Consequently, $\det H_\Omega = 0$ marks a degeneracy of the equilibrium equations and plays a central role in the fold analysis of Sections 3–4.

Let (x_e, y_e) be an equilibrium. Linearizing (8) about (x_e, y_e) gives

$$
\dot{\mathbf{X}} = A\mathbf{X}, \qquad \mathbf{X} = \begin{pmatrix} X \\ Y \\ \dot{X} \\ \dot{Y} \end{pmatrix},
\tag{13}
$$

with

$$
A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ \Omega_{xx} & \Omega_{xy} & 0 & 2 \\ \Omega_{xy} & \Omega_{yy} & -2 & 0 \end{pmatrix}_{(x_e, y_e)}.
\tag{14}
$$

Its characteristic polynomial is

$$
\lambda^4 + (4 - \Omega_{xx} - \Omega_{yy})\,\lambda^2 + \left(\Omega_{xx}\Omega_{yy} - \Omega_{xy}^2\right) = 0.
\tag{15}
$$

Because (15) is an even polynomial with real coefficients, the spectrum is symmetric under both $\lambda \mapsto -\lambda$ and complex conjugation. Hence the four eigenvalues occur as

$$
\{\lambda, -\lambda, \bar{\lambda}, -\bar{\lambda}\},
$$

with the usual reductions when some members coincide.

An equilibrium is spectrally unstable whenever at least one eigenvalue has positive real part. It is spectrally stable when the spectrum consists of two purely imaginary conjugate pairs,

$$
\lambda = \pm i\omega_1, \qquad \lambda = \pm i\omega_2, \qquad \omega_1, \omega_2 > 0.
$$

We use "spectrally stable" and "linearly stable" in this restricted sense throughout. This condition concerns only the linearized dynamics and does not by itself imply nonlinear or KAM stability. The centre frequencies ω_1 and ω_2 identified here provide the linear modes from which the periodic-orbit families of Section 6 are initialized.

2.3 Jacobi integral and Hill regions

The rotating-frame equations (8) admit the Jacobi integral. Multiplying the first equation by $\dot{x}$ and the second by $\dot{y}$, and then adding, gives

$$\dot{x}\ddot{x} + \dot{y}\ddot{y} = \Omega_x \dot{x} + \Omega_y \dot{y},$$

because the Coriolis terms cancel identically. Hence

$$\frac{d}{dt}\left[\frac{1}{2}\left(\dot{x}^2 + \dot{y}^2\right) - \Omega(x,y)\right] = 0.$$

We therefore adopt the standard Jacobi constant

$$C_J = 2\Omega(x,y) - \left(\dot{x}^2 + \dot{y}^2\right), \tag{16}$$

which is constant along every trajectory.

Since the kinetic term is non-negative, motion at a prescribed value of C_J is possible only where

$$2\Omega(x,y) \geq C_J. \tag{17}$$

The corresponding Hill region is

$$\mathcal{H}(C_J) = \left\{(x,y) : 2\Omega(x,y) \geq C_J\right\}, \tag{18}$$

and its boundary,

$$2\Omega(x,y) = C_J, \tag{19}$$

is the zero-velocity curve.

At an equilibrium (x_e, y_e) the rotating-frame velocity vanishes, so its critical Jacobi value is

$$C_{J,e} = 2\Omega(x_e, y_e). \tag{20}$$

These equilibrium values provide the critical levels at which the geometry of the regular Hill region can change through the opening or closing of necks. Their relation to the equilibrium branches and to a representative Hill-region topology change is examined in Section 7.

3 Numerical methods

The numerical analysis has three stages. First, the finite-precision reference configurations are placed on the central-configuration manifold and the equilibria of the infinitesimal body are located independently (Section 3.1). Second, the admissible central-configuration curves are traced by pseudo-arclength continuation while the equilibrium search is repeated at every continuation point and the resulting roots are associated into branches (Section 3.2). Third, changes in equilibrium multiplicity are localized and tested against the nondegeneracy conditions of a generic saddle-node fold (Section 3.3).

All tables and figures in Sections 4–7 are generated from the output of the same numerical implementation. We distinguish throughout between the arclength coordinate ℓ along a central-configuration continuation curve and the normalized arc parameter $s \in [0,1]$ obtained by rescaling ℓ over an individual admissible arc.

3.1 Reference configurations and equilibrium search

The reference configurations are specified only to finite precision and therefore do not lie exactly on the central-configuration manifold; their initial constraint residuals satisfy

$$6 \times 10^{-3} \lesssim |\psi| \lesssim 7 \times 10^{-2}.$$

For each configuration we hold (a, b) fixed and refine either c_1 or c_2, choosing the coordinate that requires the smaller displacement to reach

$$\psi(a, b, c_1, c_2) = 0.$$

The scalar refinement is carried out by a bracketed Brent solve. The resulting configurations satisfy

$$|\psi| < 10^{-12}.$$

For comparison, the accompanying reproducibility archive reports both the adopted single-coordinate displacement and the distance to the constrained nearest point of $\psi = 0$ in the (c_1, c_2) plane when both coordinates are allowed to vary. Every refined configuration is then checked for membership in the admissible set (5): all three mass ratios are positive, the denominators in (4) remain nonzero, and no primary collision occurs.

At a fixed admissible configuration, the equilibria (10) are located by an independent multi-start Newton search that uses no previously supplied equilibrium coordinates. For the reference configurations the initial guesses consist of a deterministic 81×81 grid over

$$[-5, 5]^2$$

together with 2000 uniformly distributed fixed-seed random starts. From each initial point we apply a damped Newton iteration to

$$F(x, y) = \begin{pmatrix} \Omega_x \\ \Omega_y \end{pmatrix} = 0,$$

using the analytic Hessian H_Ω of (12) as the Newton Jacobian. A backtracking line search is used to enforce decrease of $\|F\|$.

Because the effective potential is singular at the primary locations, initial points and Newton iterates falling within

$$r_{\min} = 10^{-3}$$

of any primary are rejected. As an additional guard against retaining solutions numerically attached to a primary singularity, any converged root within

$$10 r_{\min} = 10^{-2}$$

of a primary is discarded.

The remaining converged solutions are clustered in the (x, y) plane with tolerance 10^{-6} so that repeated convergence to the same equilibrium is counted only once. Every retained root is required to satisfy

$$\|\nabla\Omega\| < 10^{-12}. \tag{21}$$

The equilibrium count is therefore an output of the multi-start search rather than an input to it. This procedure provides a numerical search for all equilibria over the stated set of initial conditions; it is not an analytic completeness theorem. The additional searches used to verify the eleven-equilibrium regime are described in Section 4.3.1.

Because the analytic derivatives enter both the equilibrium solver and the subsequent bifurcation and stability calculations, they are independently validated against central finite differences at 25 test points chosen away from the primary singularities. The analytic gradient (9) is compared with finite differences of Ω, and H_Ω with finite differences of the analytic gradient. The maximum discrepancies are of order 10^{-9} for the gradient and 10^{-9}–10^{-10} for the Hessian; independent finite-difference evaluations of the mixed derivative Ω_{xy} also agree at this level.

At each equilibrium we compute the spectrum of the variational matrix (14). For numerical classification we define

$$\sigma = \max_j \operatorname{Re} \lambda_j$$

and use the threshold

$$\tau_\sigma = 10^{-8}.$$

An equilibrium is classified as spectrally stable when $\sigma \le \tau_\sigma$ and the computed spectrum consists of two numerically purely imaginary conjugate pairs; otherwise it is classified as unstable. On spectrally stable equilibria the two positive centre frequencies are recorded as

$$\omega_1 \ge \omega_2 > 0.$$

We also record $\det H_\Omega$, which is used as a degeneracy diagnostic in the fold calculation below, and the equilibrium Jacobi value (20).

3.2 Central-configuration continuation and branch tracking

For the continuation study, (a, b) are held fixed and we write

$$\mathbf{q} = (c_1, c_2).$$

The level set

$$\psi(\mathbf{q}) = 0$$

is traced by pseudo-arclength continuation. At an accepted point $\mathbf{q}_k$ with unit tangent $\mathbf{t}_k$, the predictor is

$$\mathbf{q}^* = \mathbf{q}_k + \Delta\ell\, \mathbf{t}_k,$$

and the corrector solves

$$\begin{cases} \psi(\mathbf{q}) = 0, \\ \mathbf{t}_k^T(\mathbf{q} - \mathbf{q}_k) - \Delta\ell = 0. \end{cases} \tag{22}$$

Newton's method is applied to (22), with $\nabla\psi$ evaluated by central differences using step $h = 10^{-7}$. The tangent is chosen orthogonal to the constraint gradient,

$$\mathbf{t} \propto \left(-\psi_{c_2}, \psi_{c_1}\right),$$

and its orientation is fixed by requiring

$$\mathbf{t}_k \cdot \mathbf{t}_{k-1} > 0.$$

This allows turning points in either c_1 or c_2 to be traversed without changing continuation parameter.

The initial continuation step is

$$\Delta\ell = 0.01.$$

On a failed correction it is halved down to a minimum of 10^{-7}; after successful corrections it is increased by a factor of 1.15, subject to the maximum value 0.01. Accepted continuation points satisfy

$$|\psi| < 10^{-13}.$$

Admissibility is checked after every successful correction. Each component is followed in both tangent directions until the admissible set is left. For the arcs encountered in the present study this loss of admissibility occurs when one of the positive mass ratios tends to zero. The corresponding boundary is localized by 40 bisections and the vanishing mass ratio is identified. Each maximal admissible arc is then reparametrized by normalized arclength

$$s \in [0, 1].$$

Continuation of already known equilibria is not sufficient to determine the equilibrium multiplicity, because a pair created at a saddle-node does not exist on the opposite side of the fold. We therefore repeat an independent multi-start equilibrium search at every accepted central-configuration point. For this stage the search set is enlarged to an 81×81 grid over

$$[-8, 8]^2,$$

1000 fixed-seed random starts, and a 35×35 local grid

$$[x_i - 0.8, x_i + 0.8] \times [y_i - 0.8, y_i + 0.8]$$

around each primary. The local searches are particularly important when a mass ratio becomes small, because equilibria associated with that primary can lie close to the corresponding singularity.

The same Newton solver and singularity guards as in Section 3.1 are used: initial points and Newton iterates within 10^{-3} of a primary are rejected, converged roots within 10^{-2} are discarded, and every retained root must satisfy (21). Roots are clustered at 10^{-6} within the equilibrium solver; the continuation driver performs an additional numerical de-duplication before assembling the root set at each sample.

For branch-dependent quantities, roots at neighbouring continuation samples are associated by predictor-based one-to-one matching in the equilibrium (x, y) plane. Suppose that an active branch has its two most recent positions $\mathbf{z}_{k-1}$ and $\mathbf{z}_k$. Its next position is predicted by linear extrapolation,

$$\mathbf{z}_{\text{pred}} = 2\mathbf{z}_k - \mathbf{z}_{k-1},$$

and its most recent inter-sample displacement is

$$\delta = \|\mathbf{z}_k - \mathbf{z}_{k-1}\|_2.$$

A candidate root $\mathbf{z}$ is eligible for association when

$$\|\mathbf{z} - \mathbf{z}_{\text{pred}}\|_2 \leq \max(0.18,\, 3\delta). \tag{23}$$

For a newly initialized branch with only one previous position, the predictor is that position and the implementation uses the fallback $\delta = 0.1$. Candidate branch–root pairs are ordered by predictor distance and assigned greedily subject to a one-to-one correspondence.

Because an accepted association may still be geometrically weak, a separate reliability criterion is imposed. A match is classified as reliable only when

$$\|\mathbf{z} - \mathbf{z}_{\text{pred}}\|_2 \leq \max(0.25,\, 2\delta). \tag{24}$$

Wider accepted matches, together with rapidly escaping roots near a mass-vanishing boundary, are flagged unreliable and excluded from branch-dependent stability, frequency, Jacobi, and periodic-orbit claims. The equilibrium count itself is determined independently by the multi-start root search and therefore does not depend on branch association. The same adaptive branch assignment is used for all branch-dependent diagnostics, figures, and periodic-orbit parent selections reported below.

3.3 Fold localization and verification

Whenever the independent equilibrium search detects a change in multiplicity between neighbouring continuation samples, the transition is first localized by seven bisections in the central-configuration parameter, with the multi-start equilibrium search repeated at every midpoint. The equilibria present on the higher-count side but absent on the lower-count side identify the candidate merging pair.

The midpoint of this pair and the corresponding bracketed primary configuration are then used to initialize a Newton solve of the augmented system

$$\mathcal{G}(x, y, c_1, c_2) = \begin{pmatrix} \Omega_x \\ \Omega_y \\ \det H_\Omega \\ \psi \end{pmatrix} = \mathbf{0}. \tag{25}$$

Equation (25) consists of four equations in the four unknowns (x, y, c_1, c_2), with (a, b) fixed. It is solved by damped Newton iteration using a finite-difference Jacobian with step 10^{-6}, and an augmented-system residual not exceeding 10^{-11} is required. Thus the critical equilibrium is simultaneously placed on the central-configuration manifold and at a degeneracy of the equilibrium map.

Vanishing of $\det H_\Omega$ alone does not establish a generic saddle-node. Let μ_0 denote the eigenvalue of H_Ω having the smallest absolute value at the critical point, let $\mu_\perp$ denote the other eigenvalue, and let $\mathbf{v}$ be a unit vector spanning the null direction,

$$H_\Omega \mathbf{v} = 0.$$

A simple fold requires

$$\mu_0 = 0, \qquad \mu_\perp \neq 0,$$

together with nonzero transversality and quadratic-nondegeneracy coefficients

$$\alpha = \mathbf{v}^T \left. \frac{d}{d\ell} F(\mathbf{z}_c; \mathbf{p}(\ell)) \right|_{\ell=\ell_c}, \qquad \beta = \frac{1}{2} D^3\Omega(\mathbf{z}_c)[\mathbf{v}, \mathbf{v}, \mathbf{v}], \tag{26}$$

where $\mathbf{z}_c = (x_c, y_c)$ is the critical equilibrium and $\mathbf{p}(\ell)$ denotes the primary configuration along the central-configuration curve. In the parameter derivative defining α, the equilibrium position $\mathbf{z}_c$ is held fixed while the primary geometry and masses vary consistently along the CC curve through (4). Both α and β are evaluated by centred finite differences.

For

$$\alpha \neq 0, \qquad \beta \neq 0,$$

Lyapunov–Schmidt reduction of the equilibrium equations onto the null direction gives the local fold normal form

$$\alpha(\ell - \ell_c) + \beta u^2 + \cdots = 0,$$

and hence

$$u_{\pm} \sim \pm\sqrt{-\frac{\alpha}{\beta}(\ell - \ell_c)}.$$

The separation d of the merging pair must therefore satisfy

$$d \propto |\ell - \ell_c|^{1/2}. \tag{27}$$

We test this prediction independently by resolving the merging pair on the side of the fold on which it exists and fitting

$$d \propto |\ell - \ell_c|^{\gamma}$$

on the innermost resolved points in log–log coordinates. Since s is a smooth monotone rescaling of ℓ on each admissible arc, the exponent is unchanged when expressed in terms of $|s - s_c|$.

Finally, the variational spectrum is evaluated along every reliably tracked equilibrium branch. Where two centre frequencies are present, the ratio ω_1/ω_2 is compared with the low-order rational ratios examined in Section 5, using the numerical proximity tolerance 0.04. These near-resonance flags are indicators for the subsequent periodic-orbit analysis; they are not treated as evidence of exact nonlinear resonance.

4 Equilibrium bifurcation structure

We now apply the numerical framework of Section 3 to determine how the restricted equilibria change as the underlying four-primary central configuration is varied. We first establish the four reference configurations used as the computational baseline (Section 4.1). We then examine the common Case-1/Case-2 slice and determine the component structure of its admissible central configurations (Section 4.2), map the equilibrium multiplicity along the resulting admissible arcs and resolve the eleven-equilibrium regime (Section 4.3), and finally verify every interior multiplicity change as a generic saddle-node fold (Section 4.4).

4.1 Baseline configurations

The four reference configurations are first refined onto the central-configuration manifold using the procedure of Section 3.1. At each refined configuration the mass ratios are recomputed from (4), admissibility is checked, and the restricted equilibria are located independently by the multi-start search. The results are summarized in Table 1. Every retained equilibrium satisfies

$$\|\nabla\Omega\| < 10^{-12}.$$

The resulting equilibrium counts are

$$N_{\text{eq}} = 5,\ 9,\ 5,\ 5 \qquad \text{for Cases 1–4, respectively.} \tag{28}$$

Thus the finite-precision reference geometries can be placed consistently on the central-configuration manifold without altering their equilibrium multiplicities.

One feature of this baseline will become important in Section 5. Case 1 contains an outer equilibrium near

$$(x, y) \simeq (2.67,\ 2.62)$$

whose linear spectrum consists of two purely imaginary conjugate pairs. It is therefore spectrally stable, whereas the remaining equilibria of the four reference configurations are spectrally unstable. The complete equilibrium coordinates, refinement displacements, and per-root numerical diagnostics are provided in the accompanying reproducibility archive.

Table 1: The four representative non-symmetric configurations used as the computational baseline. The approximate geometry is refined onto the central-configuration manifold $\psi = 0$ while holding (a, b) fixed, after which the mass ratios are recomputed. N_{eq} is the equilibrium count returned by the independent multi-start search, with every retained root satisfying $\|\nabla\Omega\| < 10^{-12}$.

Case	(a,b)	$(c_1,c_2)_{\text{approx}}$	$(c_1,c_2)_{\text{refined}}$	(ρ_0,ρ_1,ρ_2)	N_{eq}
Case 1	$(0.9,-0.1)$	$(-0.61,0.71)$	$(-0.610000,0.715655)$	$(49.654,3.0981,1.1533)$	5
Case 2	$(0.9,-0.1)$	$(0.52,-0.88)$	$(0.517714,-0.880000)$	$(4.2776,0.95681,1.2772)$	9
Case 3	$(1.5,0.4)$	$(-0.96,0.99999)$	$(-0.944942,0.999990)$	$(3.0082,0.15809,0.43247)$	5
Case 4	$(1.5,0.4)$	$(1.05,0.13)$	$(1.050584,0.130000)$	$(4.3119,0.98292,0.72214)$	5

4.2 Distinct central-configuration components on the fixed slice

Cases 1 and 2 share the same values

$$(a, b) = (0.9, -0.1),$$

while possessing different equilibrium multiplicities, $N_{\text{eq}} = 5$ and $N_{\text{eq}} = 9$, respectively. Their common fixed-(a, b) slice therefore provides a natural controlled setting in which to ask whether the two configurations belong to the same admissible central-configuration family and how the restricted equilibrium structure changes along that family. The slice is selected directly from the reference configurations, before the continuation analysis; the disconnected component structure and the multiplicity cascade reported below are outcomes of the continuation rather than criteria used to choose the slice.

We consequently continue

$$\psi(0.9, -0.1, c_1, c_2) = 0 \tag{29}$$

by the pseudo-arclength method of Section 3.2. The continuation reveals two qualitatively different zero-set branches containing the reference configurations: an open branch through Case 1 and a closed loop through Case 2 (Figure 1). Restricting these branches to positive mass ratios produces the two admissible arcs used throughout the remainder of the paper.

The closed character of the Case-2 branch is obtained directly from the pseudo-arclength trace: continuation around the branch in the (c_1, c_2) plane returns to the starting neighbourhood and completes the loop. No vanishing of

$$\nabla_{(c_1,c_2)}\psi$$

is encountered along the accepted continuation points. Thus, within the resolution of the numerical trace, the Case-2 loop is a regular level-set component. At a regular point the implicit-function theorem gives a locally unique smooth zero-set curve, so another branch cannot join or leave the loop there. The Case-1 reference configuration lies instead on the separately traced open branch.

The numerical continuation therefore supports the conclusion that Cases 1 and 2 lie on distinct connected components of the regular level set $\psi = 0$ on the investigated fixed-(a, b) slice. This is deliberately a slice-specific numerical statement, not a global theorem for the complete four-parameter central-configuration set. In particular, the present calculation does not exclude the possibility that a path connecting the two configurations exists when a and b are also allowed to vary.

The consequence for the restricted problem is important. The five- and nine-equilibrium reference configurations are not two points on a single fixed-(a, b) continuation family. Their different multiplicities therefore cannot be interpreted as evidence of a bifurcation sequence directly joining Case 1 to Case 2. Equilibrium multiplicity must instead be analysed separately on the two components.

On the branch through Case 1, the maximal admissible arc containing the reference configuration extends from a boundary where $\rho_0 \to 0$ to one where $\rho_2 \to 0$. On the branch through Case 2, the corresponding admissible arc extends from $\rho_1 \to 0$ to $\rho_0 \to 0$. Each maximal admissible arc is parametrized below by normalized arclength

$$s \in [0, 1].$$

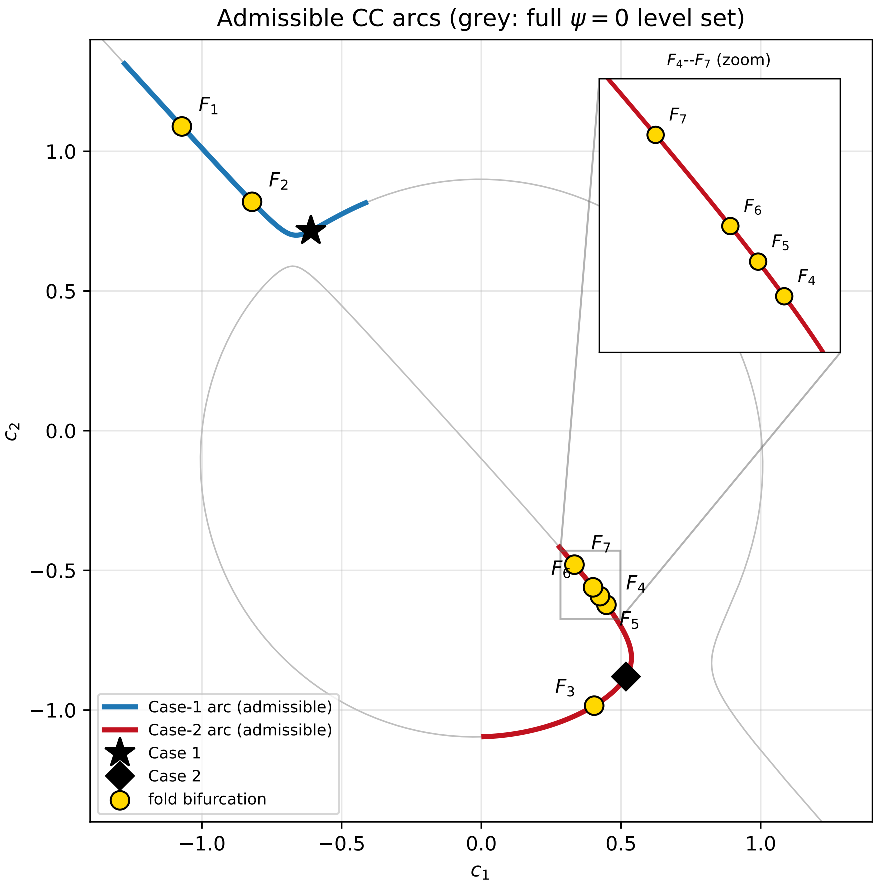

Figure 1: Central-configuration branches and admissible arcs in the (c_1, c_2) plane at $(a, b) = (0.9, -0.1)$. Grey shows the numerically represented $\psi = 0$ branches in the plotted domain; blue and red denote the admissible arcs containing Cases 1 and 2, respectively. Case 1 is marked by a star and Case 2 by a diamond. Gold circles mark the seven verified interior saddle-node folds F_1–F_7; the inset resolves the closely spaced folds F_4–F_7. The two reference configurations lie on distinct fixed-slice components, and each admissible arc terminates where a primary mass ratio vanishes.

4.3 Fold cascades and the eleven-equilibrium regime

The equilibrium multiplicity cannot be determined by continuing only the branches already known at a reference configuration, because a pair born at a saddle-node does not exist on the opposite side of its fold. We therefore use the independent multi-start search described in Section 3.2 at every accepted central-configuration sample.

Table 2: Verified changes in equilibrium multiplicity along the two admissible central-configuration arcs. Interior count changes are generic saddle-node folds located from $\Omega_x = \Omega_y = \det H_\Omega = \psi = 0$ and satisfying the nondegeneracy conditions of Section 3.3. The coordinates $(c_1, c_2)_c$ describe the critical *primary configuration*; (α, β) are the transversality and quadratic-nondegeneracy coefficients of (26). The Case-1 $5 \to 4$ event is a mass-vanishing boundary degeneration rather than an interior fold.

Arc	N_{eq}	Fold	s_c	$(c_1, c_2)_c$	$(\rho_0, \rho_1, \rho_2)_c$	(α, β)	Type
Case-1	$5 \to 7$	F_1	0.261	$(-1.0717, 1.0890)$	$(0.507, 0.993, 0.965)$	$(-2.72, 22.4)$	saddle-node
Case-1	$7 \to 5$	F_2	0.583	$(-0.8206, 0.8193)$	$(3.95, 1.78, 1.65)$	$(16.3, 42.6)$	saddle-node
Case-1	$5 \to 4$	n/a	0.974	n/a	n/a	n/a	$\rho_2 \to 0$ boundary
Case-2	$7 \to 9$	F_3	0.366	$(0.4040, -0.9842)$	$(8.9, 0.667, 1.27)$	$(-0.615, 0.0566)$	saddle-node
Case-2	$9 \to 11$	F_4	0.768	$(0.4478, -0.6231)$	$(0.71, 2.19, 2.28)$	$(-18.8, 43.4)$	saddle-node
Case-2	$11 \to 9$	F_5	0.804	$(0.4246, -0.5923)$	$(0.576, 2.49, 2.57)$	$(1.18, 0.231)$	saddle-node
Case-2	$9 \to 7$	F_6	0.839	$(0.3998, -0.5606)$	$(0.456, 2.87, 2.94)$	$(-6.03, -48.8)$	saddle-node
Case-2	$7 \to 5$	F_7	0.929	$(0.3333, -0.4795)$	$(0.189, 4.3, 4.35)$	$(0.506, 0.173)$	saddle-node

The resulting multiplicity function $N_{\text{eq}}(s)$ is shown in Figure 2. Along the two admissible arcs we obtain

$$\begin{aligned} &\text{Case-1 arc:} \qquad 5 \to 7 \to 5 \to 4, \\ &\text{Case-2 arc:} \qquad 7 \to 9 \to 11 \to 9 \to 7 \to 5. \end{aligned} \tag{30}$$

The seven interior changes all have magnitude two and are subsequently verified as generic saddle-node folds. They are denoted $F_1, \ldots, F_7$ in Figures 1–2 and listed in Table 2.

The final $5 \to 4$ change on the Case-1 arc is qualitatively different. It occurs near

$$s \simeq 0.974$$

as $\rho_2 \to 0$, when the underlying positive-mass primary configuration approaches the boundary of the admissible set and an associated equilibrium recedes to large distance. This is therefore a mass-vanishing parameter-boundary degeneration, not an interior saddle-node bifurcation. More generally, the admissible CC arcs terminate when one of the primary mass ratios vanishes, but such arc termination must not be conflated with an interior bifurcation of the restricted equilibrium equations.

The local primary searches of Section 3.2 are especially important near these limits, where equilibria associated with a small primary mass can lie close to the corresponding singularity. The multiplicities in (30) are those obtained after applying the augmented global-plus-local search throughout the arcs.

4.3.1 The eleven-equilibrium regime

The most pronounced multiplicity increase occurs on the Case-2 arc. The $9 \to 11$ fold F_4 at

$$s_c \simeq 0.768$$

and the $11 \to 9$ fold F_5 at

$$s_c \simeq 0.804$$

bound a narrow interval in which eleven distinct restricted equilibria are detected. The two folds occur at different primary configurations (Table 2) and at spatially separated restricted equilibria. They are therefore two distinct saddle-node events rather than a single higher-codimension degeneracy.

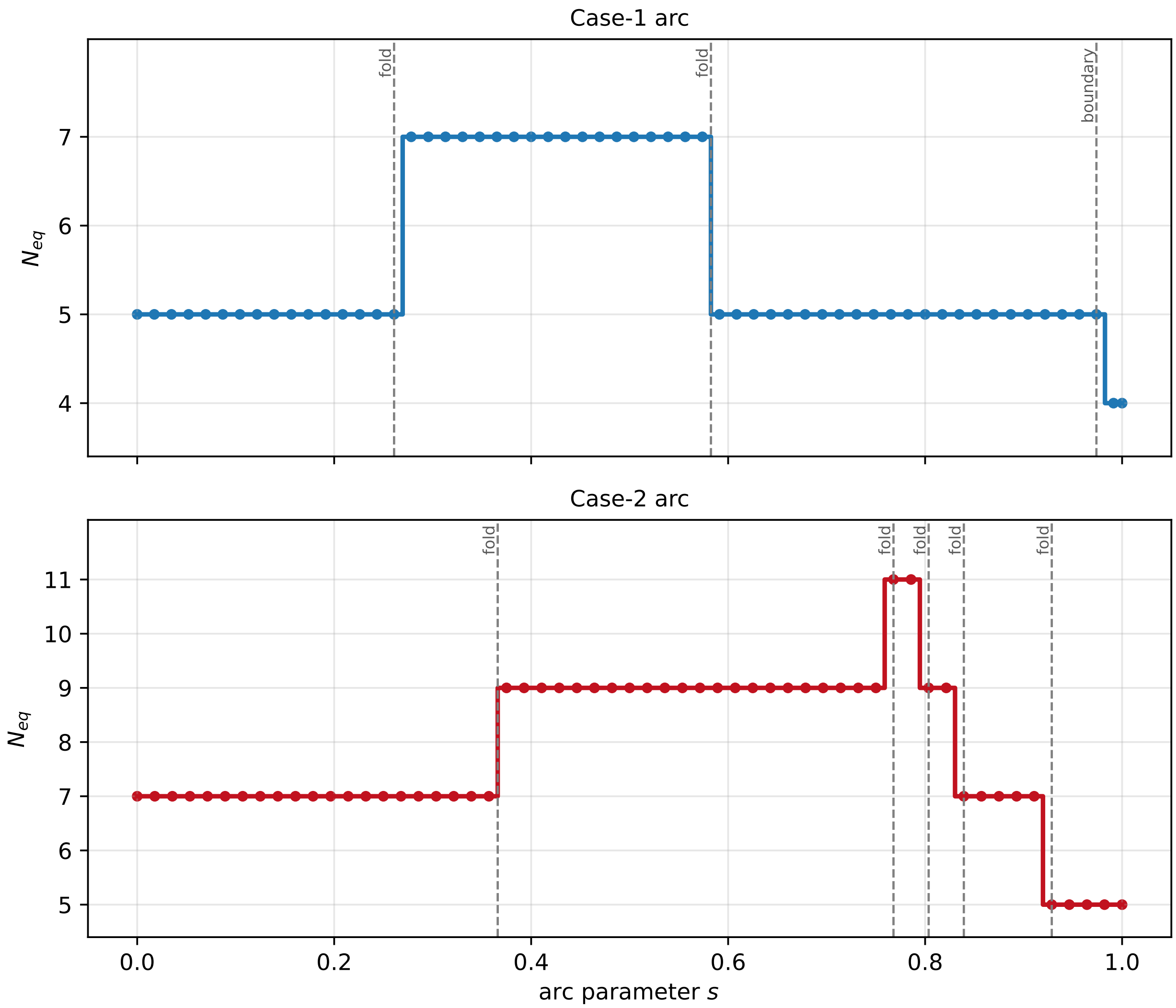

Figure 2: Equilibrium multiplicity $N_{\mathrm{eq}}(s)$ along the admissible arcs. Top: Case-1 arc; bottom: Case-2 arc. Dashed vertical lines mark the verified saddle-node folds F_1–F_7. The right-most Case-1 transition is the mass-vanishing event $\rho_2 \to 0$ and is not an interior fold.

Tracking the individual branches through this interval reveals a feature that is not visible from the multiplicity alone. At F_4, a new pair, denoted B_a, is created near the critical restricted equilibrium

$$(x, y) \simeq (-0.38,\, 0.31).$$

At the representative point

$$s \simeq 0.786$$

inside the eleven-equilibrium interval, its two members lie approximately at

$$(-0.439,\, 0.377), \qquad (-0.301,\, 0.218).$$

The pair destroyed at the subsequent $11 \to 9$ fold F_5 is not B_a. Instead, F_5 annihilates a pre-existing pair, denoted B_b, in the lower part of the equilibrium configuration near

$$(x, y) \simeq (0.53,\, -1.70).$$

Immediately before the fold its two members lie approximately at

$$(0.26,\, -1.74), \qquad (0.74,\, -1.64).$$

Branch tracking shows that this pair was created earlier at the $7 \to 9$ fold F_3.

The newly created pair B_a therefore survives the $11 \to 9$ transition and remains present throughout the following nine-equilibrium interval. It is annihilated only at F_6, the subsequent $9 \to 7$ fold near

$$(x, y) \simeq (-0.26,\, 0.12).$$

Consequently,

$$9 \to 11 \to 9$$

is not a simple creation and subsequent destruction of the same equilibrium pair. The two nine-equilibrium intervals on either side of the eleven-equilibrium window have the same multiplicity but different branch populations. The eleven-equilibrium interval therefore exhibits a genuine branch exchange.

As a robustness check, the equilibrium search was repeated at the representative point

$$s \simeq 0.786, \qquad (c_1, c_2) \simeq (0.432, -0.602),$$

inside the eleven-equilibrium interval. The re-search uses the same 81×81 deterministic grid over $[-8, 8]^2$ and the same 35×35 local grid around each primary as the main continuation analysis, but increases the number of fixed-seed random starts from 1000 to 1500 (seed 20260808). It again returns eleven distinct equilibria. Every retained root satisfies

$$\|\nabla\Omega\| < 10^{-12}.$$

The minimum pairwise separation among all eleven roots at this configuration is approximately

$$0.175,$$

while the separation of the two members of the newly created pair B_a is approximately

$$0.21.$$

We therefore report eleven as the number of distinct equilibria detected under the stated numerical search procedure. This is a numerically supported multiplicity on the investigated parameter interval; it is not an analytic upper bound on the number of real equilibria of the general restricted five-body problem.

Figure 3 shows representative equilibrium geometries through the complete Case-2 sequence

$$7 \to 9 \to 11 \to 9 \to 7 \to 5,$$

with B_a and B_b identified explicitly. The radial branch representation in Figure 4 gives the complementary continuation view and makes the branch exchange visible even where the scalar multiplicity alone cannot distinguish branch identity.

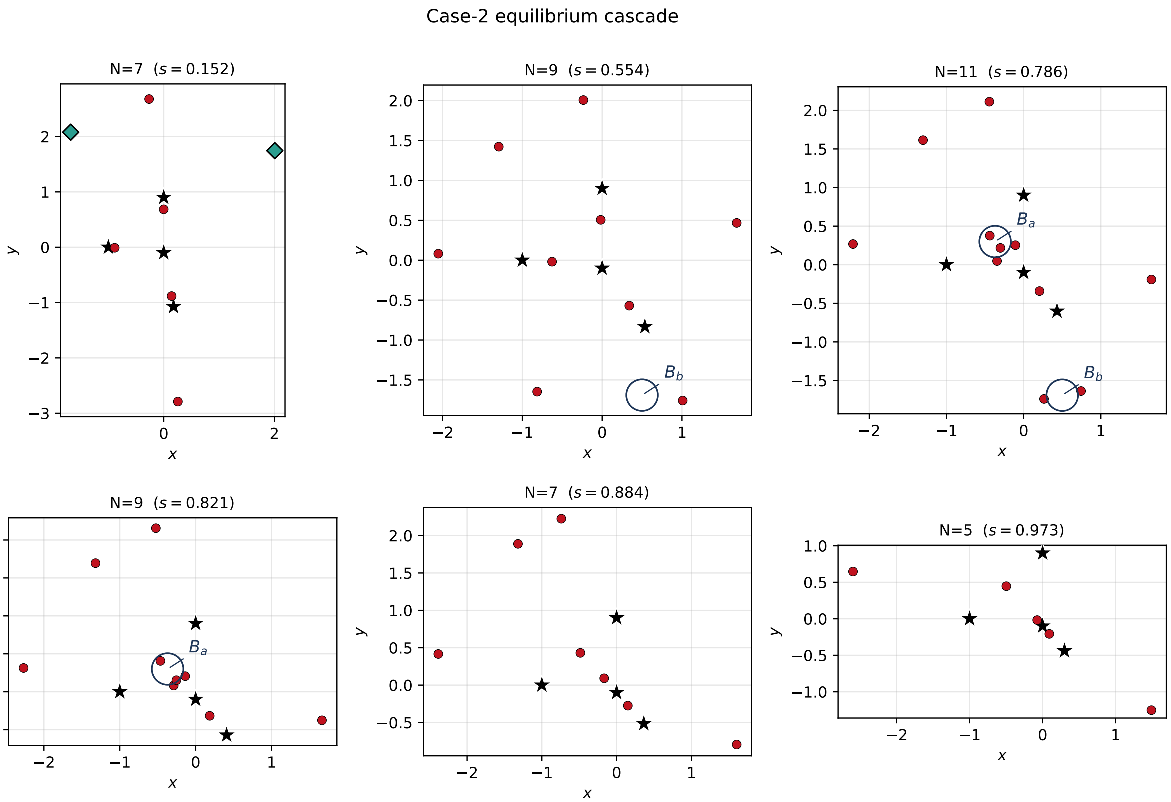

Figure 3: Equilibrium geometry at representative points of the Case-2 cascade $7 \to 9 \to 11 \to 9 \to 7 \to 5$. Stars denote primaries, red circles unstable equilibria, and green diamonds spectrally stable equilibria. The branch exchange in the eleven-equilibrium interval is annotated explicitly: B_a, created at F_4, persists into the following $N_{\mathrm{eq}} = 9$ interval, whereas the pre-existing pair B_b is destroyed at F_5.

The branch identities shown in Figure 4 are those obtained from the adaptive predictor-based tracker of Section 3.2. The same branch assignment is used for the stability, centre-frequency, Jacobi, and periodic-orbit analyses that follow. Ordinary equilibrium branches are therefore shown without individual labels, while the dynamically important pairs B_a and B_b are highlighted explicitly. Large-radius excursions near the ends of the admissible arcs correspond to equilibria receding as a primary mass ratio approaches zero.

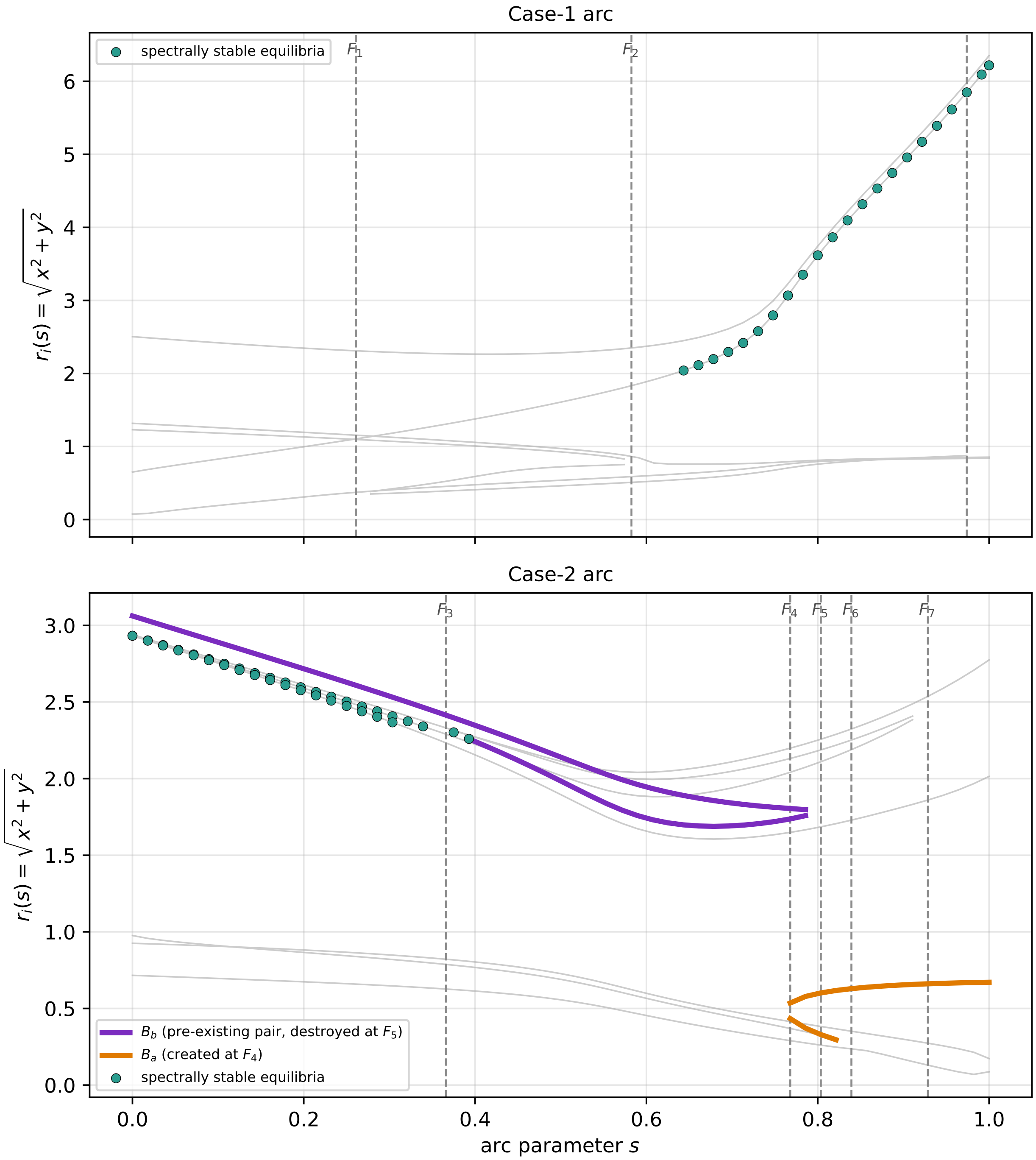

Figure 4: Radial equilibrium coordinate $r_i(s) = \sqrt{x_i(s)^2 + y_i(s)^2}$ along the two admissible arcs. Ordinary branches are shown in muted grey and dashed vertical lines mark folds F_1–F_7. Spectrally stable equilibria are highlighted as described in Section 5. On the Case-2 arc, B_a is the pair created at F_4, whereas B_b is the pre-existing pair destroyed at F_5, directly displaying the branch exchange in the eleven-equilibrium interval. Large-radius excursions near the arc ends correspond to equilibria receding at mass-vanishing limits.

4.4 Saddle-node verification and scaling

The seven interior multiplicity changes in (30) are tested using the fold criteria of Section 3.3. For every transition, the augmented system

$$\Omega_x = \Omega_y = \det H_\Omega = \psi = 0 \tag{31}$$

converges to a distinct critical point with augmented-system residual not exceeding 10^{-11}.

At each critical point the Hessian has one numerically vanishing eigenvalue, denoted μ_0, and one nonzero transverse eigenvalue, denoted $\mu_\perp$. The corresponding transversality and quadratic-nondegeneracy coefficients satisfy

$$\alpha \neq 0, \qquad \beta \neq 0$$

for all seven folds, with the computed (α, β) values listed in Table 2. Thus every interior count change satisfies the numerical conditions for a simple generic saddle-node fold.

The local branch geometry provides an independent verification. For each fold the merging pair is resolved on the side of the critical parameter on which both equilibria exist. The two roots approach one another and meet at the numerically determined fold location, as illustrated for representative cases in Figure 5.

We further fit their separation to

$$d \propto |\ell - \ell_c|^\gamma \tag{32}$$

using the innermost resolved points. Across the seven folds the fitted exponents satisfy

$$0.44 \leq \gamma \leq 0.52,$$

consistent with the Lyapunov–Schmidt prediction

$$\gamma = \frac{1}{2}$$

derived in Section 3.3. Since the normalized coordinate s is a smooth monotone rescaling of ℓ on each admissible arc, the same local exponent is obtained when the distance to the critical point is expressed using $|s - s_c|$.

The agreement of four independent diagnostics (the observed $\Delta N_{\mathrm{eq}} = \pm 2$ count changes, convergence of the augmented degeneracy system, satisfaction of the fold nondegeneracy conditions, and the square-root pair-separation law) provides mutually consistent evidence that $F_1, \ldots, F_7$ are generic saddle-node bifurcations of the restricted equilibrium equations.

5 Stability and centre modes

We now examine the linear stability of the equilibrium branches obtained in Section 4 and identify the centre modes from which the periodic-orbit families of Section 6 are continued. The spectral criterion and numerical threshold have already been defined in Sections 2.2 and 3.1; here we focus on the resulting stable intervals, the evolution of their centre frequencies, and the occurrence of low-order near-commensurabilities.

5.1 Spectral stability along the equilibrium branches

For each reliably tracked equilibrium branch we compute

$$\sigma(s) = \max_j \operatorname{Re} \lambda_j,$$

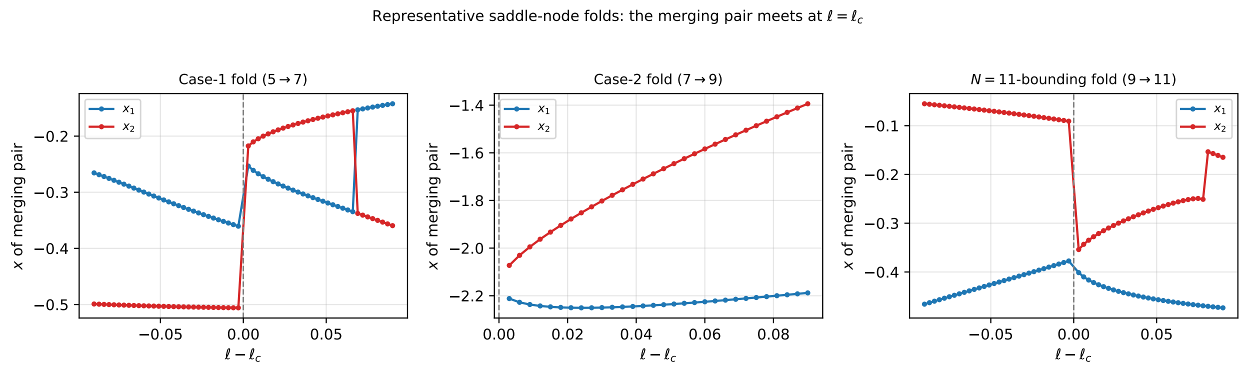

Figure 5: Three representative saddle-node folds (a Case-1 fold, an ordinary Case-2 fold, and the $9 \to 11$ fold bounding the eleven-equilibrium interval): the x-coordinates of the merging pair meet at $\ell = \ell_c$ (dashed).

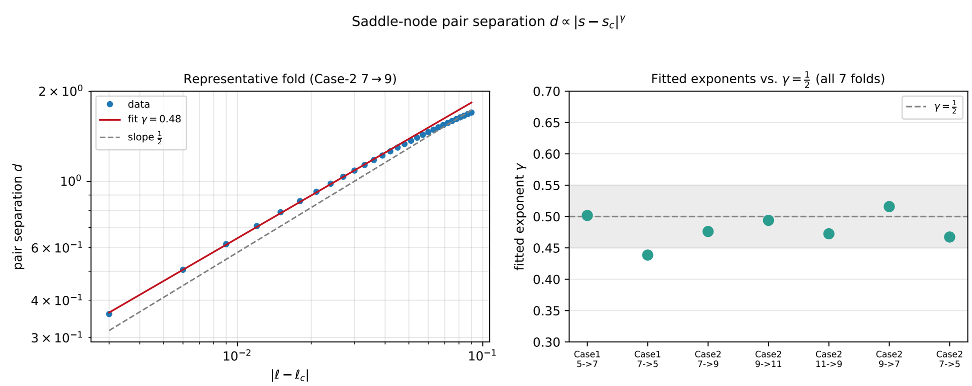

Figure 6: Saddle-node pair-separation scaling. Left: a representative fold (Case-2 $7 \to 9$) showing the merging-pair separation d against $|\ell - \ell_c|$ on log–log axes, with the fitted slope and the reference slope $\frac{1}{2}$. Right: the fitted exponents γ of all seven folds against the theoretical value $\gamma = \frac{1}{2}$ (shaded band $[0.45, 0.55]$); all lie in $[0.44, 0.52]$.

where λ_j are the eigenvalues of the rotating-frame variational matrix (14). As in Section 3.1, an equilibrium is classified as spectrally stable when

$$\sigma \leq 10^{-8}$$

and the spectrum consists numerically of two purely imaginary conjugate pairs. This is a statement about linear spectral stability only and does not imply nonlinear or KAM stability.

The continuation shows that spectral stability is not confined to isolated reference configurations. On the Case-1 arc, one equilibrium branch is reliably tracked as spectrally stable over

$$s \in [0.643,\, 0.835].$$

This branch contains the outer stable equilibrium identified in Section 4.1. Stable spectral classifications are also obtained farther toward the mass-vanishing end of the arc, but branch tracking in this rapidly escaping region is flagged unreliable by the criterion of Section 3.2; those points are therefore not used for branch-dependent conclusions. At the refined Case-1 configuration its position is approximately

$$(x, y) = (2.671,\, 2.622),$$

and its spectrum is

$$\lambda = \pm 0.9632\, i, \qquad \lambda = \pm 0.2984\, i, \tag{33}$$

with real parts at the level of machine precision.

The Case-2 arc contains a richer stability structure. Two distinct equilibrium branches are spectrally stable over substantial and overlapping parts of the arc (Figure 7). One remains stable from the beginning of the admissible arc to approximately

$$s = 0.339,$$

and passes through

$$(x, y) \simeq (-1.70,\, 1.96)$$

near $s = 0.196$. A second stable branch persists to approximately

$$s = 0.304$$

and passes through

$$(x, y) \simeq (2.06,\, 1.55)$$

at nearly the same value of s. Thus a single primary configuration can support more than one spectrally stable libration point. Isolated single-sample classifications farther along the Case-2 arc are not used to define additional stable intervals.

The pairs genuinely created at the seven interior saddle-node folds are spectrally unstable at the first resolved points away from their folds. In particular, the pair B_a created at F_4 in the eleven-equilibrium regime acquires nonzero real eigenvalues immediately after its creation. The stable branches described above are instead pre-existing equilibrium branches whose spectra become purely imaginary over finite parameter intervals.

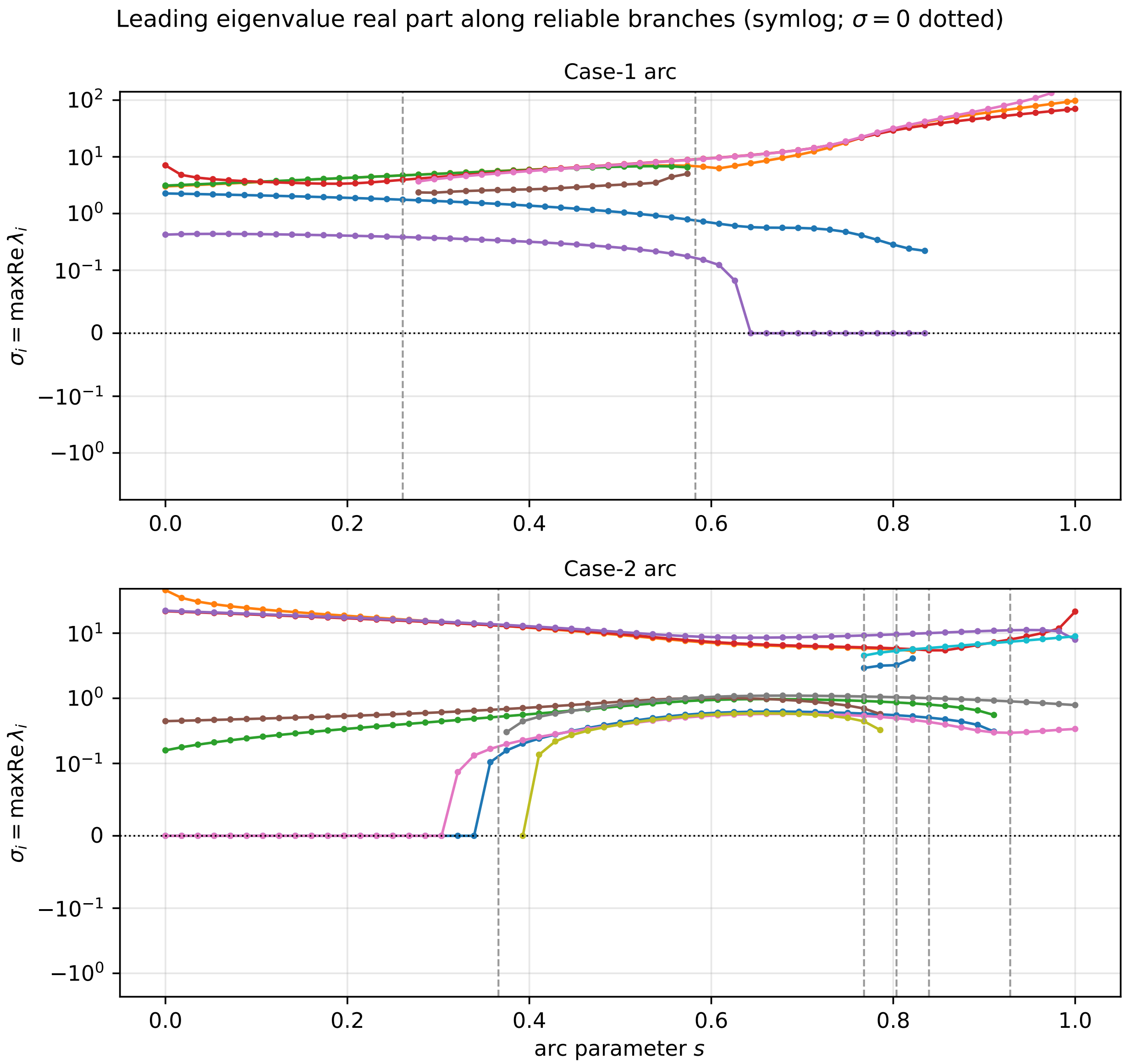

Figure 7: Leading eigenvalue real part $\sigma_i(s)$ along the reliably tracked equilibrium branches. Dashed vertical lines mark the saddle-node folds. On the Case-1 arc one branch is reliably tracked as spectrally stable over $s \simeq [0.643, 0.835]$. On the Case-2 arc two distinct branches are spectrally stable, extending approximately to $s = 0.339$ and $s = 0.304$, respectively. Spectral stability means that both eigenvalue pairs are purely imaginary to the numerical threshold of Section 3.1.

5.2 Centre frequencies and low-order near-commensurabilities

At a doubly-elliptic equilibrium the spectrum has the form

$$\lambda = \pm i\omega_1, \qquad \lambda = \pm i\omega_2, \qquad \omega_1 \geq \omega_2 > 0.$$

We follow both positive centre frequencies along each reliably stable branch and examine the ratio

$$R(s) = \frac{\omega_1(s)}{\omega_2(s)}.$$

A point is flagged as a low-order near-commensurability when $R(s)$ lies within 0.04 of one of

$$1, \quad \frac{3}{2}, \quad 2, \quad \frac{5}{2}, \quad 3, \quad 4.$$

These flags are numerical proximity indicators only. They do not establish an exact resonance or a nonlinear resonant bifurcation.

On the reliably tracked Case-1 stable branch, one low-order near-commensurability is resolved (Figure 8). Near

$$s \simeq 0.678$$

the equilibrium is approximately

$$(x, y) \simeq (2.062, -0.756),$$

with

$$\omega_1 \simeq 0.8509, \qquad \omega_2 \simeq 0.5645, \qquad \frac{\omega_1}{\omega_2} \simeq 1.5075,$$

placing it within 0.0075 of the 3:2 ratio.

The two stable Case-2 branches exhibit different frequency-ratio structures. On the branch passing through

$$(x, y) \simeq (-1.70,\ 1.96)$$

at $s \simeq 0.196$, the centre frequencies are

$$\omega_1 \simeq 0.9172, \qquad \omega_2 \simeq 0.3721,$$

so that

$$\frac{\omega_1}{\omega_2} \simeq 2.465. \tag{34}$$

This is within approximately 0.035 of the 5:2 ratio. No 2:1 near-commensurability occurs on this branch.

The second stable Case-2 branch passes successively near several low-order ratios:

$$\begin{aligned}
s \simeq 0.018 \quad &: \quad \omega_1/\omega_2 \simeq 3.995 \quad (4{:}1),\\
s \simeq 0.125 \quad &: \quad \omega_1/\omega_2 \simeq 2.510 \quad (5{:}2),\\
s \simeq 0.196 \quad &: \quad \omega_1/\omega_2 \simeq 1.983 \quad (2{:}1),\\
s \simeq 0.268 \quad &: \quad \omega_1/\omega_2 \simeq 1.488 \quad (3{:}2).
\end{aligned}$$

At the genuine Case-2 2:1 near-commensurability the equilibrium lies near

$$(x, y) \simeq (2.06,\ 1.55),$$

with

$$\omega_1 \simeq 0.8933, \qquad \omega_2 \simeq 0.4506.$$

The occurrence of two different stable equilibria at essentially the same arc parameter $s \simeq 0.196$ is important: one lies near 5:2 and the other near 2:1. The arc parameter alone therefore does not uniquely identify the dynamical state; branch identity must also be retained.

The periodic-orbit dataset of Section 6 samples selected members of this near-commensurate structure rather than attempting an exhaustive resonance survey. Families 7–8 are continued from the Case-1 equilibrium near the 3:2 ratio. Families 9–10 are continued from the first Case-2 stable branch, namely the equilibrium near

$$(-1.70, 1.96)$$

whose frequency ratio is approximately 2.465 and is therefore associated with the 5:2 near-commensurability. The distinct Case-2 equilibrium near the 2:1 ratio is not used as a parent in the present periodic-orbit dataset.

5.3 Lyapunov-centre interpretation

The centre modes identified above provide the local seeds for the periodic-orbit families of Section 6. It is therefore useful to state precisely when the Lyapunov centre theorem applies.

Let $\pm i\omega_k$ be the simple centre pair selected for continuation at a parent equilibrium. The local Lyapunov family exists provided no other eigenvalue of the variational matrix is an integer multiple

$$\pm m i\omega_k, \qquad m = 2, 3, \ldots$$

of that selected frequency. This condition is satisfied at every parent used for the ten periodic-orbit families computed in Section 6.

For the saddle×centre parents of families 5–6, the companion eigenvalue pair is real. It therefore cannot equal an integer multiple of the purely imaginary centre frequency. The hyperbolic direction makes the resulting periodic families unstable, but it does not obstruct their local existence as Lyapunov families.

For the doubly-elliptic parents, proximity to a rational frequency ratio must be distinguished from exact resonance. In particular, the Case-1 parent of families 7–8 has

$$\omega_1/\omega_2 \simeq 1.5075,$$

near but not equal to 3:2, while the Case-2 parent of families 9–10 has

$$\omega_1/\omega_2 \simeq 2.465,$$

near but not equal to 5:2. Neither ratio violates the exact non-resonance hypothesis of the theorem. The same distinction applies to the other near-commensurabilities identified above: numerical proximity to 2:1 or 4:1 is not an exact integer resonance.

Accordingly, all ten families computed in Section 6 are Lyapunov families in the local theorem sense. What differs among them is the stability type of the parent equilibrium and the proximity of its centre frequencies to low-order commensurabilities. Whether a continued family subsequently encounters a resonant or period-doubling bifurcation is a global continuation question and must be decided from its Floquet multipliers rather than inferred from the linear frequency ratio alone.

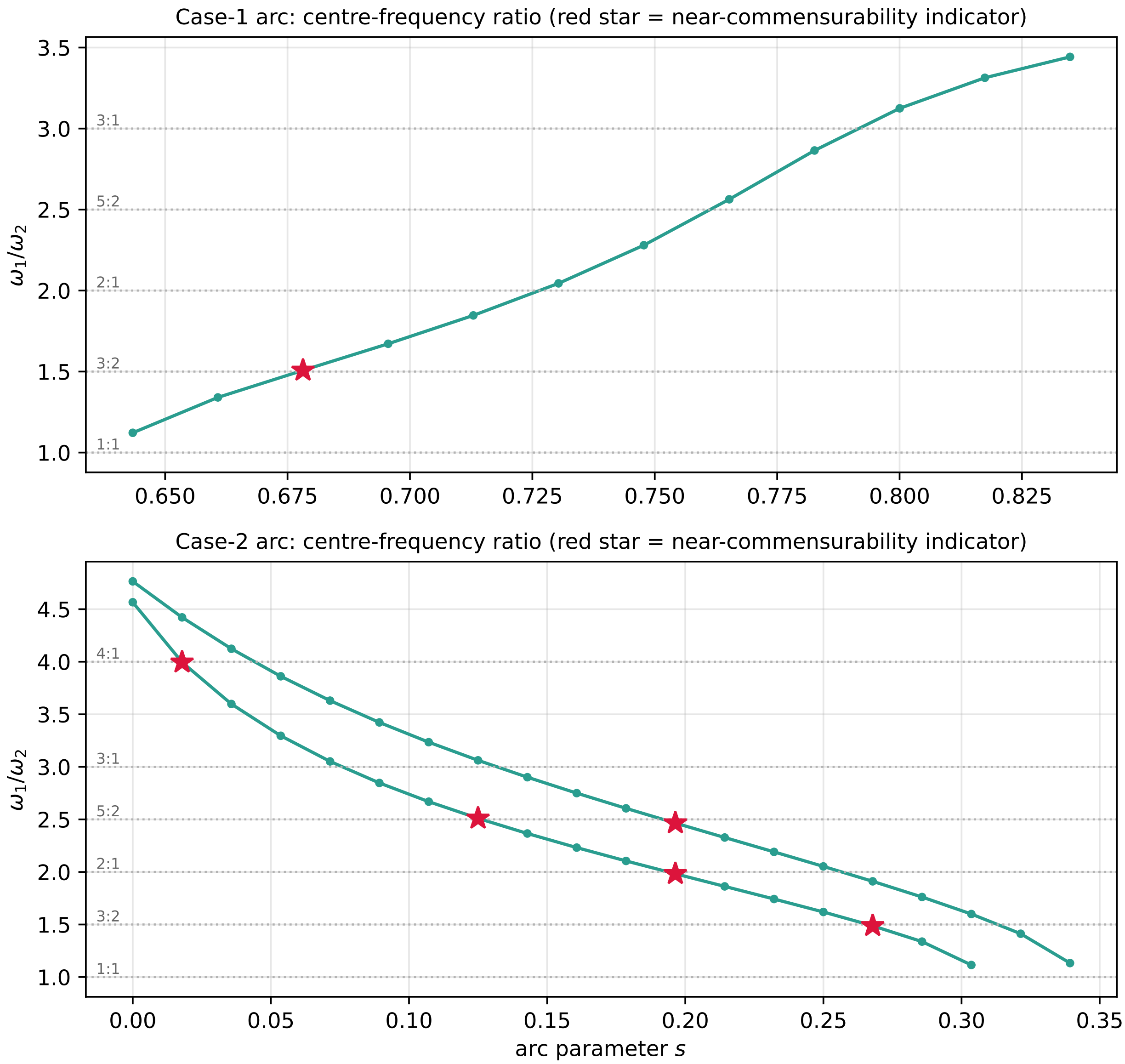

Figure 8: Centre-frequency ratio ω_1/ω_2 along the reliably resolved spectrally stable branches. Dotted horizontal lines mark selected low-order rational ratios, and highlighted points indicate proximity within the tolerance 0.04. These points are near-commensurability indicators only and do not establish exact resonance. The Case-2 panel contains two distinct stable equilibrium branches; at $s \simeq 0.196$ one lies near 5:2 and the other near 2:1.

Table 3: Computed periodic-orbit families. Each family is a Lyapunov family seeded from a simple centre mode of a parent equilibrium and continued in amplitude; the Lyapunov centre theorem applies at every parent (the exact non-resonance condition holds, including at the saddle×centre parents of families 5–6 and the near-resonant parents of families 7–10; see Section 5.3). What differs is the parent stability type and proximity to resonance, not theoretical standing. 'mode' is the centre frequency ω; N the number of corrected members; the C_J and T ranges are over the family; 'max closure' is the largest periodicity residual $\|u(T) - u_0\|$ attained over the family's members (all $< 10^{-10}$). Floquet class is by the stability index ν ($|\nu| < 2$ stable).

Fam	parent (arc, eq.)	ω	N	C_J range	T range	max closure	Floquet / note
1	Case-1 stable, $(2.78, 0.25)$	0.925	22	$[23.89, 24.03]$	$[6.78, 6.79]$	5.3×10^{-11}	Floquet stable
2	Case-1 stable, $(2.78, 0.25)$	0.406	20	$[24.03, 24.04]$	$[15.48, 15.52]$	8.1×10^{-11}	Floquet stable
3	Case-2 stable, $(1.99, 1.79)$	0.922	23	$[21.34, 21.79]$	$[6.68, 6.82]$	3.5×10^{-11}	Floquet stable
4	Case-2 stable, $(1.99, 1.79)$	0.390	20	$[21.79, 21.80]$	$[16.13, 16.51]$	4.1×10^{-11}	Floquet stable
5	Case-1 sdl×ctr, $(-0.97, 0.51)$	3.738	18	$[13.33, 13.43]$	$[1.68, 1.72]$	7.6×10^{-11}	real unstable
6	Case-2 sdl×ctr, $(0.35, -0.64)$	7.435	15	$[24.28, 24.29]$	$[0.85, 0.85]$	8.6×10^{-11}	real unstable
7	Case-1 3:2 reg., $(2.06, -0.76)$	0.851	20	$[15.23, 15.25]$	$[7.38, 7.38]$	6.0×10^{-11}	Floquet stable
8	Case-1 3:2 reg., $(2.06, -0.76)$	0.564	21	$[15.25, 15.29]$	$[11.13, 11.39]$	1.7×10^{-11}	Floquet stable; $\nu \to -2$
9	Case-2 5:2 reg., $(-1.70, 1.96)$	0.917	21	$[20.28, 20.36]$	$[6.79, 6.85]$	9.0×10^{-11}	Floquet stable
10	Case-2 5:2 reg., $(-1.70, 1.96)$	0.372	18	$[20.36, 20.36]$	$[16.89, 17.17]$	2.6×10^{-11}	Floquet stable; $\nu \to -2$

6 Periodic-orbit families

The centre modes identified in Section 5.2 provide local seeds for Lyapunov families of periodic orbits. We compute ten such families, chosen to sample doubly-elliptic equilibria, saddle×centre equilibria, and selected low-order near-commensurability regions. This section describes the shooting formulation and continuation procedure, classifies the corrected orbits by their Floquet multipliers, and examines the approach of two families to the period-doubling threshold. The computed families are summarized in Table 3.

6.1 Shooting formulation and linear seeding

Write the rotating-frame state as

$$\mathbf{u} = (x, y, \dot{x}, \dot{y})^T$$

and denote by $\phi_T(\mathbf{u}_0)$ the time-T flow of (8). A periodic orbit satisfies

$$\phi_T(\mathbf{u}_0) - \mathbf{u}_0 = \mathbf{0}. \tag{35}$$

Let

$$L_j = (x^*, y^*, 0, 0)^T$$

be a parent equilibrium with a simple centre eigenvalue $+i\omega_k$ and corresponding complex eigenvector $\mathbf{w}_k$ of the variational matrix (14). The small-amplitude linear motion provides the initial approximation

$$\mathbf{u}(t) = L_j + \varepsilon \operatorname{Re}\left(e^{i\omega_k t}\mathbf{w}_k\right).$$

Using the real part of the selected eigenvector as the initial direction, we take

$$\mathbf{u}_0^{(0)} = L_j + \varepsilon \widehat{\mathbf{e}}_k, \qquad \widehat{\mathbf{e}}_k = \frac{\operatorname{Re}\mathbf{w}_k}{\|\operatorname{Re}\mathbf{w}_k\|}, \qquad T_0 = \frac{2\pi}{\omega_k}. \tag{36}$$

At a doubly-elliptic equilibrium the two simple centre modes are seeded separately, producing two local Lyapunov families.

The periodicity equations alone do not isolate a unique member of a one-parameter family because the autonomous flow also admits arbitrary time translation along a periodic orbit. We therefore supplement the four periodicity equations with a phase condition and an amplitude condition,

$$\widehat{\mathbf{e}}_k^T f(\mathbf{u}_0) = 0, \qquad \widehat{\mathbf{e}}_k^T(\mathbf{u}_0 - L_j) = a, \tag{37}$$

where f denotes the rotating-frame vector field and a is the prescribed continuation amplitude.

The complete correction residual is consequently

$$\mathcal{R}(\mathbf{u}_0, T) = \begin{pmatrix} \phi_T(\mathbf{u}_0) - \mathbf{u}_0 \\ \widehat{\mathbf{e}}_k^T f(\mathbf{u}_0) \\ \widehat{\mathbf{e}}_k^T(\mathbf{u}_0 - L_j) - a \end{pmatrix}. \tag{38}$$

The state-transition matrix $\Phi(t)$ is integrated simultaneously from

$$\dot{\Phi} = A(\mathbf{u}(t))\Phi, \qquad \Phi(0) = I,$$

and the monodromy matrix is

$$M = \Phi(T).$$

The derivatives of the periodicity residual are

$$D_{\mathbf{u}_0}\big[\phi_T(\mathbf{u}_0) - \mathbf{u}_0\big] = M - I, \qquad D_T\big[\phi_T(\mathbf{u}_0) - \mathbf{u}_0\big] = f(\phi_T(\mathbf{u}_0)).$$

Thus the 6×5 correction Jacobian is

$$J = \begin{pmatrix} M - I & f(\phi_T(\mathbf{u}_0)) \\ \widehat{\mathbf{e}}_k^T A(\mathbf{u}_0) & 0 \\ \widehat{\mathbf{e}}_k^T & 0 \end{pmatrix}, \tag{39}$$

and each Newton correction is obtained from the least-squares problem

$$\delta X = \arg\min_{\delta} \|J\delta + \mathcal{R}\|_2.$$

The phase and amplitude constraints remove the two neutral directions associated with time translation and displacement along the local one-parameter family. The resulting overdetermined formulation retains all four periodicity equations symmetrically rather than eliminating a state coordinate through the Jacobi integral.

6.2 Continuation and numerical validation

The state, period, and state-transition matrix are integrated with DOP853 using

$$\mathrm{rtol} = \mathrm{atol} = 10^{-12}.$$

Newton corrections are capped according to

$$\|\delta\mathbf{u}_0\| \le 0.5, \qquad |\delta T| \le 0.5T,$$

with $T > 0$ enforced throughout.

A corrected orbit is accepted only when

$$\|\phi_T(\mathbf{u}_0) - \mathbf{u}_0\|_2 < 10^{-10}, \qquad |\widehat{\mathbf{e}}_k^T f(\mathbf{u}_0)| < 10^{-10}, \qquad |\widehat{\mathbf{e}}_k^T(\mathbf{u}_0 - L_j) - a| < 10^{-12}. \tag{40}$$

Across the 198 accepted periodic solutions the observed maxima are

$$\|\phi_T(\mathbf{u}_0) - \mathbf{u}_0\|_2 \leq 9.0 \times 10^{-11},$$
$$|\widehat{\mathbf{e}}_k^T f(\mathbf{u}_0)| \leq 4.7 \times 10^{-12}, \qquad |\widehat{\mathbf{e}}_k^T(\mathbf{u}_0 - L_j) - a| \leq 3.9 \times 10^{-16}.$$

The periodicity residual therefore provides a direct numerical certificate that every orbit reported below closes to the stated tolerance.

We continue each family using the natural amplitude parameter

$$a = \widehat{\mathbf{e}}_k^T(\mathbf{u}_0 - L_j),$$

with the prescribed amplitudes forming the geometric sequence

$$a_0 = 2 \times 10^{-3}, \qquad a_{n+1} = 1.35\, a_n. \tag{41}$$

Each new member is predicted from its predecessor and then corrected by the shooting system above, with at most 40 Newton iterations. For geometric interpretation we also record

$$A = \max_t \sqrt{(x(t) - x^*)^2 + (y(t) - y^*)^2}.$$

Several safeguards prevent continuation onto spurious or unrelated solutions. Correction is terminated after an integration failure caused by a near collision; solutions with

$$T < 0.4T_0$$

are rejected to prevent collapse toward the trivial $T \to 0$ solution; and the family is terminated if the geometric amplitude more than doubles between consecutive members. Continuation also stops when the orbit approaches a primary within 0.05 or when

$$A > 2.5.$$

Because the continuation parameter is the prescribed amplitude a, these computations determine only the portion of each family reachable while a remains a useful monotone local coordinate. We therefore make no claim that the computed curves are maximal periodic-orbit families. Continuation through a turning point in a would require pseudo-arclength continuation in the extended periodic-orbit system, which is not attempted here.

The ten parent modes were chosen to sample distinct dynamical situations. Families 1–4 use both centre modes of one doubly-elliptic equilibrium on each admissible arc. Families 5–6 use centre modes of selected saddle×centre equilibria. Families 7–8 are seeded from the Case-1 equilibrium near the 3:2 near-commensurability identified in Section 5.2, and families 9–10 from the distinct Case-2 equilibrium near the 5:2 near-commensurability. As established in Section 5.2, the latter parent is

$$(x, y) \simeq (-1.70, 1.96), \qquad \omega_1/\omega_2 \simeq 2.465,$$

and is not the separate Case-2 equilibrium lying near the 2:1 ratio.

For every family the smallest corrected members approach the intended linear mode:

$$\mathbf{u}_0(a) \to L_j, \qquad T(a) \to \frac{2\pi}{\omega_k} \qquad \text{as } a \to 0.$$

For example,

$$\frac{2\pi}{0.925} = 6.793, \qquad \frac{2\pi}{0.406} = 15.48, \qquad \frac{2\pi}{0.922} = 6.815,$$

in agreement with the smallest computed periods of families 1–3 to the reported precision.

6.3 Floquet stability

The linear stability of each corrected periodic orbit is determined from its monodromy matrix $M = \Phi(T)$. The Hamiltonian structure of the rotating-frame problem implies reciprocal symmetry of the Floquet multipliers. For a periodic orbit of this two-degree-of-freedom autonomous Hamiltonian system there is also a trivial multiplier pair at $+1$, associated with the flow and energy-family directions. Numerically this pair is recovered to within a few times 10^{-6}.

After removal of the trivial pair, one nontrivial reciprocal multiplier pair remains. If its members are denoted λ and λ^{-1}, we use the stability index

$$\nu = \lambda + \lambda^{-1}. \tag{42}$$

For the planar problem ν is real. The nontrivial multipliers lie on the unit circle when

$$|\nu| < 2$$

and form a real reciprocal pair when

$$|\nu| > 2.$$

The limiting values

$$\nu = +2 \qquad \text{and} \qquad \nu = -2$$

correspond respectively to a $+1$ collision and a period-doubling (-1) collision of the nontrivial multipliers.

We use "Floquet stable" only to mean that the nontrivial multiplier pair lies on the unit circle to the stated numerical tolerance; it does not imply nonlinear stability of the periodic orbit. Since only one nontrivial reciprocal pair remains in the planar problem, a Hamiltonian collision between two distinct nontrivial multiplier pairs cannot occur; the relevant local stability boundaries are therefore the values $\nu = \pm 2$.

Although DOP853 is not a symplectic integrator, the computed monodromy matrices preserve the expected Hamiltonian structure to good numerical accuracy. Over the 198 accepted orbits,

$$|\det M - 1| \le 1.5 \times 10^{-8}, \qquad |\lambda_1 \lambda_2 - 1| \le 1.5 \times 10^{-8},$$

with median values near 10^{-13}, while the trivial pair remains within 5×10^{-6} of $+1$ (median $\sim 3 \times 10^{-8}$). These diagnostics provide an independent numerical check on the Floquet classification.

6.4 Computed families and near-period-doubling behaviour

The ten families comprise 198 accepted periodic solutions. Their parent equilibria, centre frequencies, Jacobi and period ranges, closure residuals, and Floquet classifications are summarized in Table 3.

Before turning to the individual families, it is useful to fix their geometric character. Each parent equilibrium is a libration point of the rotating four-primary configuration, and the associated Lyapunov orbits are small oscillations *about that point* rather than revolutions about any individual primary. This is illustrated in Figure 9. For the Case-2 parent of family 3 the Hessian of Ω is positive definite, with eigenvalues 0.0436 and 2.9554, so the parent is a strict local minimum of the rotating-frame effective potential. It is also the numerically identified global minimum of 2Ω for this primary configuration, with $2\Omega_{\min} \simeq 21.7867$. The computed family members satisfy $C_J \le 2\Omega_{\min}$; consequently the corresponding numerical Hill set contains no forbidden region and is the whole plane. The nested family is therefore organized by the local potential minimum rather than by confinement within a closed zero-velocity boundary.

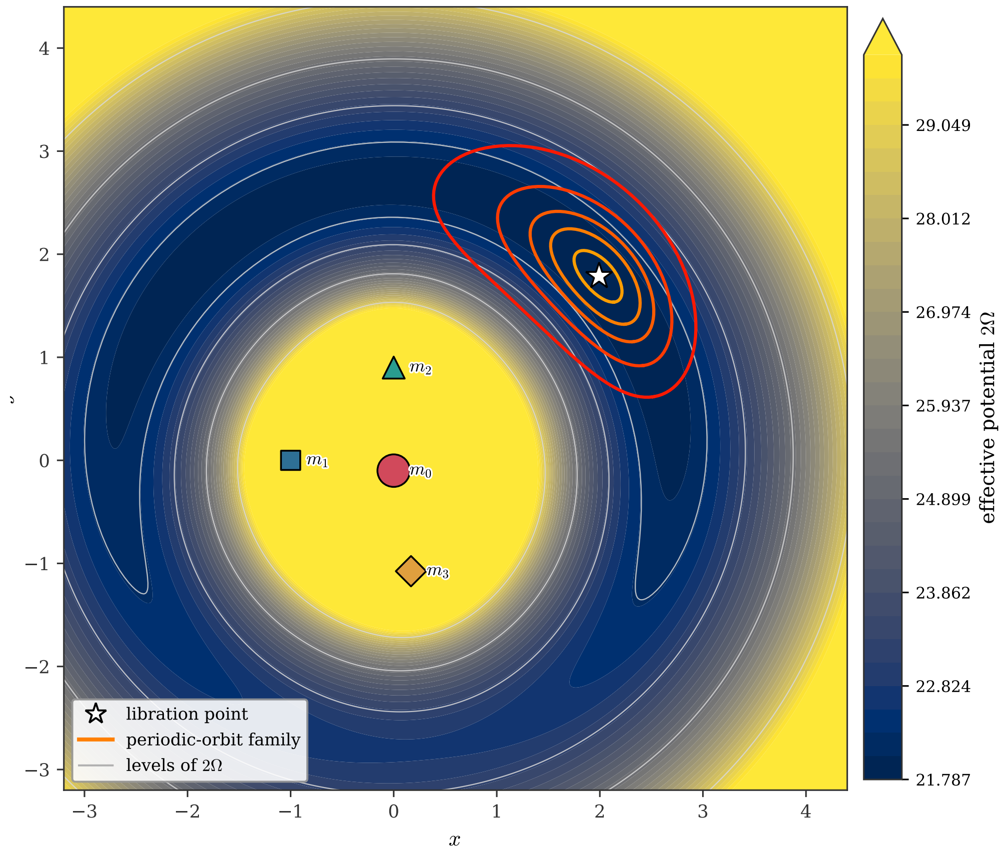

Figure 9: Effective-potential landscape for the Case-2 configuration supporting periodic-orbit family 3. Filled contours show $2\Omega(x, y)$, with the colour range clipped at large values for visualization; $2\Omega \to +\infty$ at the primary singularities and at large distance. Thin grey curves are representative level curves of 2Ω at 22.29, 23.29, 25.29, and 28.79. The four primaries are marked by m_0–m_3, and the star denotes the doubly-elliptic parent equilibrium. The Hessian of Ω at this point has eigenvalues 0.0436 and 2.9554, so the parent is a strict local minimum of the effective potential. Successive members of the associated Lyapunov family are shown from small to larger amplitude, illustrating libration about the equilibrium of the rotating four-primary configuration rather than orbital motion about any individual primary.

Families 1–4 originate from the selected doubly-elliptic equilibria on the two admissible arcs. Both centre modes of each parent produce corrected Lyapunov families whose nontrivial Floquet multipliers remain on the unit circle throughout the computed intervals. Figures 10 and 11 show representative small, intermediate, and large members. The smallest solutions reproduce the corresponding linear centre modes, while larger members deform substantially without losing Floquet stability over the continuation range. Figure 12 shows the period, geometric amplitude, and Floquet index of all ten families against C_J.

Families 5–6 demonstrate the contrasting behaviour of saddle×centre parents. Their selected centre modes satisfy the Lyapunov-centre non-resonance condition discussed in Section 5.2, so genuine local periodic families exist despite the hyperbolic companion direction. Their nontrivial Floquet multipliers, however, form a strongly real reciprocal pair, giving

$$|\nu| \sim 10^3$$

over the computed portions. These families are therefore real unstable. A representative member is shown in Figure 13.

Families 7–10 sample the near-commensurate doubly-elliptic regions. Families 7–8 originate near the Case-1 3:2 ratio, while families 9–10 originate near the Case-2 5:2 ratio. These ratios are near-commensurabilities rather than exact resonances at the parent equilibria, so their local existence as Lyapunov families is unaffected.

Two of these families approach the period-doubling boundary $\nu = -2$ as their amplitudes increase. Fine continuation near the minimum of

$$|\nu + 2|$$

shows that family 8 remains approximately

$$2 \times 10^{-3}$$

away from the threshold, whereas family 10 approaches much more closely, reaching

$$\min_a |\nu + 2| \simeq 1.0 \times 10^{-7}, \qquad \nu \simeq -1.99999990, \qquad A \simeq 0.28. \tag{43}$$

The index subsequently moves away from -2 rather than crossing it (Figure 14).

The approach was re-examined with an ultra-fine amplitude sweep (amplitude step 8×10^{-4}) around the minimum, which returns the same closest approach, $\min_a |\nu + 2| \simeq 1.0 \times 10^{-7}$ at $\nu \simeq -1.99999990$, with the index turning back rather than crossing. All orbits are integrated at rtol = atol = 10^{-12} (Section 6.2), and the monodromy diagnostics of Section 6.3 confirm that the multiplier computation remains reliable at this near-degenerate point. We therefore find no period-doubling crossing over the computed continuation interval. In particular, the calculation provides no evidence for a local period-doubled branch bifurcating from family 10 within that interval.

A corresponding scan of the computed families for crossings of the selected low-order multiplier conditions

$$\nu = 2, \qquad \nu = 0, \qquad \nu = -1, \qquad \nu = -2$$

finds none over the continuation ranges investigated. This is a negative result restricted to the computed families and amplitudes; it is not a claim that secondary resonant or period-doubled periodic solutions cannot exist elsewhere in phase or parameter space.

Finally, four distinct notions must remain separate. Spectral stability of an equilibrium refers to the eigenvalues of (14); Floquet stability of a periodic orbit refers to its monodromy multipliers;

numerical boundedness over a finite integration interval establishes neither; and nonlinear or KAM stability is not proved in this work. Every periodic orbit reported here satisfies the closure conditions (40), and every periodic stability label is a Floquet statement.

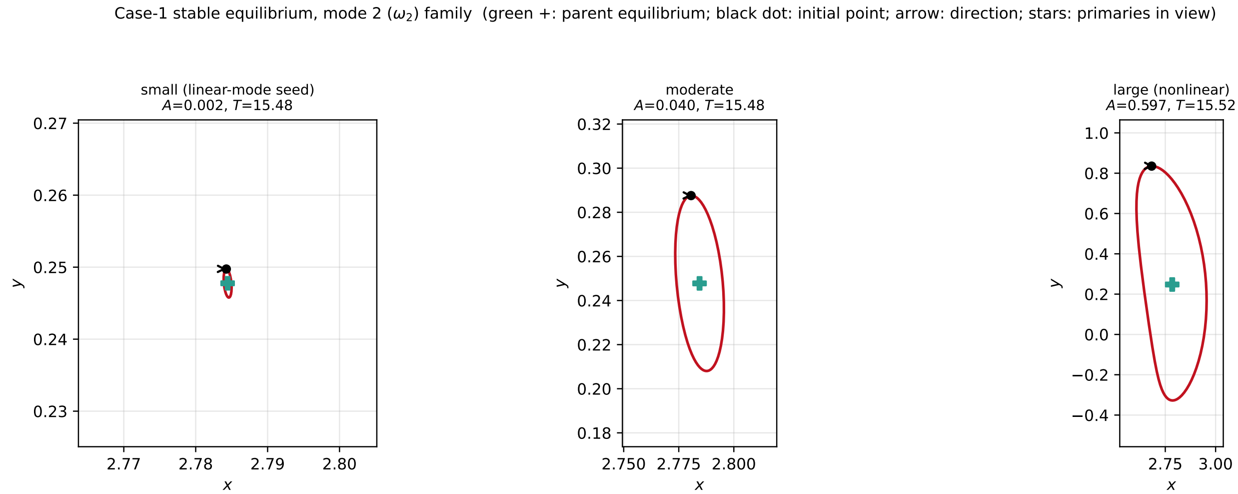

Figure 10: Stable Lyapunov family from the Case-1 arc equilibrium (mode ω_2): small (linear-mode seed), moderate, and large members. Green +: parent equilibrium; black dot: initial point; arrow: direction; stars: primaries in view. Floquet stable throughout.

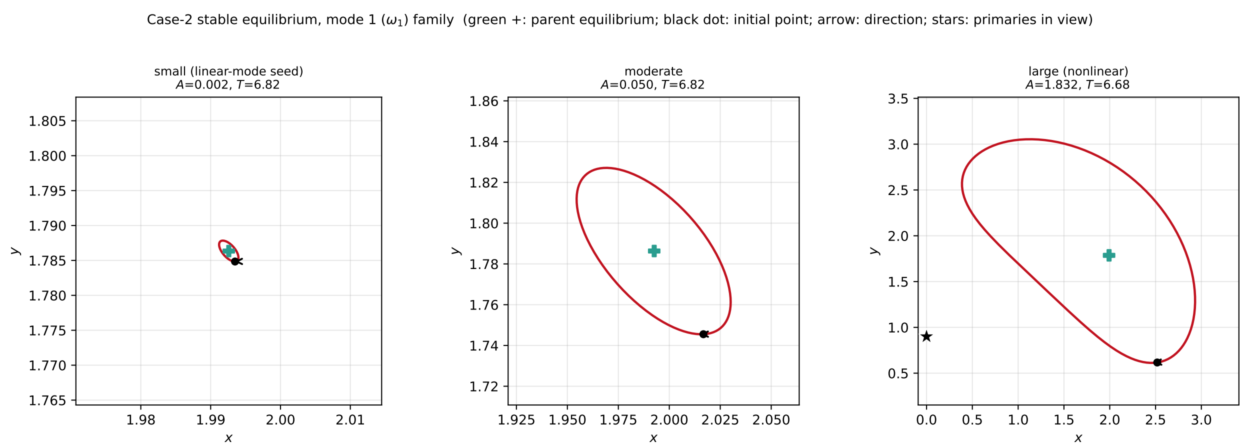

Figure 11: Stable Lyapunov family from the Case-2 arc equilibrium (mode ω_1). Floquet stable throughout.

7 Jacobi levels and Hill-region topology

The Jacobi integral provides the geometric link between the equilibrium bifurcations of Section 4 and the periodic dynamics of Section 6. We first follow the critical Jacobi values associated with the equilibrium branches, then resolve one representative change in Hill-region connectivity in the eleven-equilibrium regime, and finally place selected periodic families within their corresponding accessible regions.

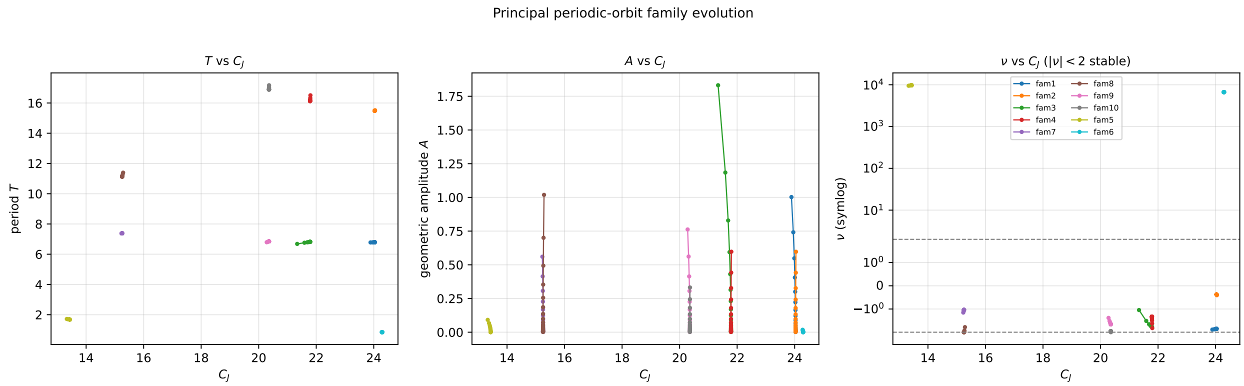

Figure 12: Family evolution: period T, geometric amplitude A, and Floquet index ν (symlog) against C_J. Real-unstable families (5, 6) sit at $\nu \sim 10^3$; all others have $|\nu| < 2$ over their computed range.

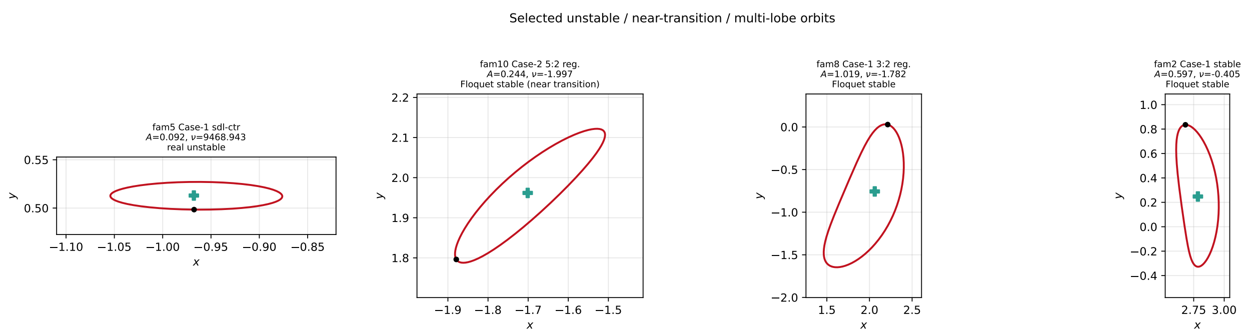

Figure 13: Representative periodic orbits: a real-unstable orbit from a saddle×centre parent (family 5), a near-period-doubling member of the Case-2 5:2 family (family 10), a multi-lobed member of the Case-1 3:2 family (family 8), and a large Floquet-stable orbit (family 2). Titles give the geometric amplitude, Floquet index ν, and Floquet class.

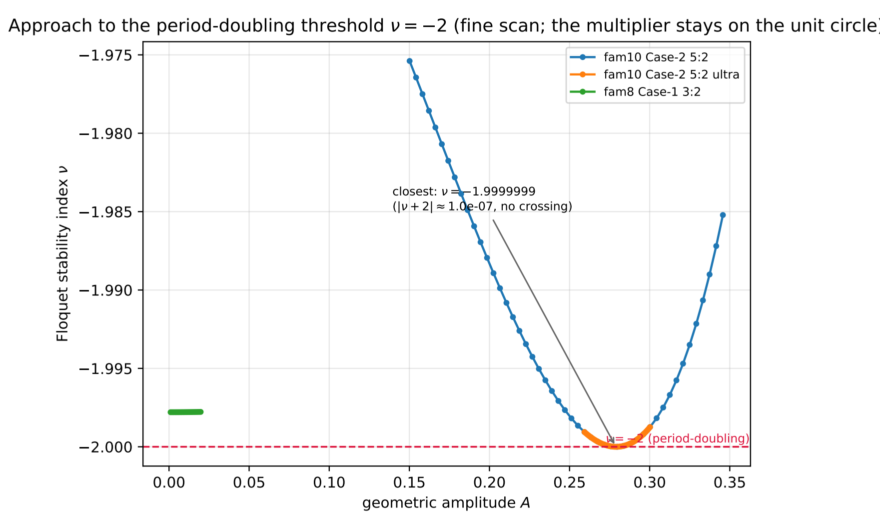

Figure 14: Fine scan of the Floquet index toward $\nu = -2$. Family 10 approaches the threshold to within $\sim 10^{-7}$ and returns; the multiplier does not cross -1. No period-doubling bifurcation occurs.

7.1 Critical Jacobi levels along the equilibrium branches

Under the convention of Section 2.3, motion at Jacobi constant C_J is restricted to the Hill region

$$\mathcal{H}(C_J) = \{(x, y) : 2\Omega(x, y) \geq C_J\}\,, \tag{44}$$

whose boundary is the zero-velocity curve

$$2\Omega(x, y) = C_J.$$

Away from the primary singularities, the restricted equilibria are the finite critical points of Ω. Each equilibrium branch $L_i(s)$ therefore determines a critical Jacobi level

$$C_{J,i}(s) = 2\Omega(L_i(s))\,. \tag{45}$$

For a regular finite level set, changes in the topology of the Hill region can occur only when C_J passes through such a critical value. Depending on the local type of the critical point, the change may involve the opening or closing of a neck, the creation or loss of a bounded component, or a change in the number of holes; a critical value does not therefore imply a change in component count in every case.

Figure 15 shows $C_{J,i}(s)$ along the reliably tracked equilibrium branches. The near-primary equilibria generally occupy the upper part of the critical level ladder, whereas the more distant equilibria occur at lower levels. Branches created or destroyed at the saddle-node folds introduce, respectively remove, pairs of critical levels. At the fold itself the two equilibrium branches coalesce at the same critical point and hence at the same Jacobi value. The fold cascade of Section 4 therefore reorganizes not only the number of equilibria but also the hierarchy of finite critical energy levels available to the restricted body.

7.2 A representative connectivity change

We resolve a representative connectivity change in the eleven-equilibrium regime on the Case-2 arc, at

$$s \simeq 0.786, \qquad (c_1, c_2) \simeq (0.432, -0.602).$$

Two Jacobi levels bracketing a portion of the interior critical-level ladder are shown in the top row of Figure 16. They are chosen as

$$C_J = 11.63 \qquad \text{and} \qquad C_J = 12.83,$$

approximately ± 0.6 about the median equilibrium critical value $C_J \simeq 12.23$ at this configuration. The interval contains three finite equilibrium critical values, approximately

$$11.824, \qquad 11.924, \qquad 12.233.$$

At $C_J = 11.63$ the allowed set forms one connected component on the computational domain, whereas at $C_J = 12.83$ an inner allowed region has detached from the outer region, giving two connected components. Increasing C_J across this interval therefore produces a net neck-closing transition that separates a bounded inner component from the unbounded exterior component. Because the interval spans a small cluster of critical values, this calculation demonstrates the net connectivity change across that portion of the critical-level ladder; it does not assign the transition to one isolated critical equilibrium.

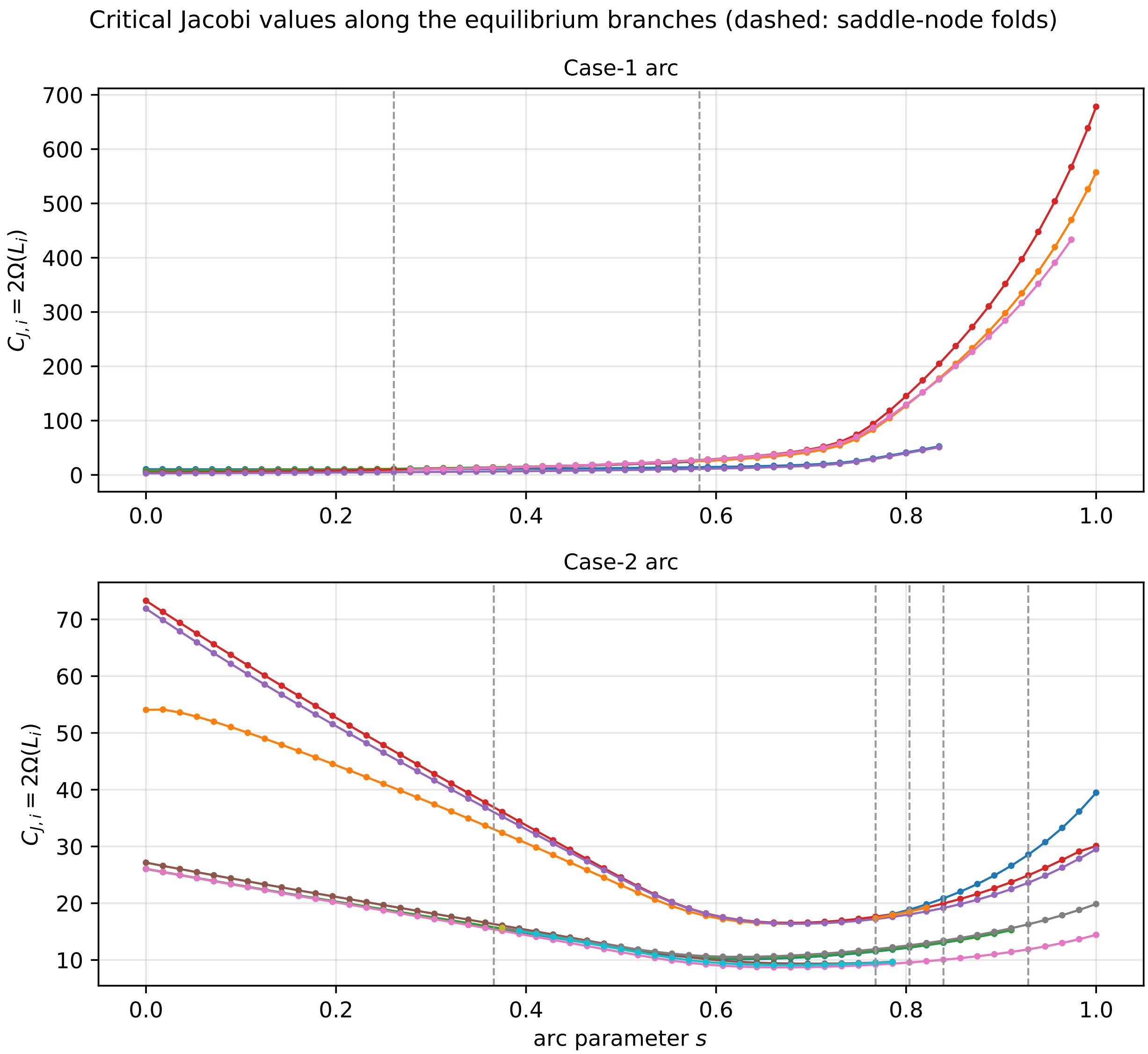

Figure 15: Critical Jacobi values $C_{J,i}(s) = 2\Omega(L_i(s))$ along the reliably tracked equilibrium branches of the two admissible arcs. Dashed vertical lines mark the saddle-node folds. Each curve gives a finite critical level at which the regular Hill-region topology may change.

The component count is evaluated numerically by sampling $2\Omega(x, y)$ on a uniform

$$400 \times 400$$

grid over

$$[-6, 6]^2,$$

forming the binary allowed set

$$2\Omega(x, y) \geq C_J,$$

and applying four-connectivity labelling. Because

$$2\Omega(x, y) \to +\infty$$

both as a primary is approached and as $\sqrt{x^2 + y^2} \to \infty$, sufficiently small punctured neighbourhoods of the primaries and the far exterior belong to the allowed region for every finite C_J. No artificial exclusion mask is introduced around the primaries in the component calculation.

The calculation was repeated on

$$300 \times 300 \qquad \text{and} \qquad 500 \times 500$$

grids. The detailed location and shape of the narrow neck are naturally resolution dependent, but the reported change from one to two connected components at the two quoted Jacobi levels is unchanged.

This calculation is intentionally representative rather than exhaustive. Figure 15 contains many other finite critical levels, and crossing them need not always alter the component count; a complete classification would require resolving the topology on both sides of every critical level. We do not attempt that census here.

7.3 Periodic families within the accessible regions

The lower panels of Figure 16 place representative Floquet-stable periodic orbits within the Hill regions corresponding to their own Jacobi constants. The Case-2 example uses family 3 at approximately

$$C_J = 21.34,$$

while the Case-1 example uses family 2 at approximately

$$C_J = 24.04.$$

In both cases the periodic orbit lies entirely inside a single connected allowed region and remains separated from the zero-velocity boundary.

For the displayed trajectories we also evaluate the Hill margin

$$\Delta_H(t) = 2\Omega(x(t), y(t)) - C_J.$$

The minimum values along the plotted family-3 and family-2 members are approximately

$$0.529 \qquad \text{and} \qquad 2.65 \times 10^{-3},$$

respectively. Both remain positive, confirming numerically that the displayed trajectories remain inside the allowed region and do not intersect the zero-velocity curve.

The examples illustrate the spatial distinction between the different dynamical regimes encountered in Section 6. Periodic librations associated with the selected outer doubly-elliptic equilibria occupy broad allowed regions away from the most intricate near-primary geometry, whereas periodic motion associated with inner saddle×centre equilibria occurs in a more strongly structured near-primary region. The Hill geometry does not determine Floquet stability, but it specifies the energetically accessible part of the plane within which each periodic family must lie.

Taken together, Figures 15 and 16 illustrate the sequence

$$\text{equilibrium bifurcation} \longrightarrow \text{critical Jacobi level} \longrightarrow \text{Hill-region topology} \longrightarrow \text{periodic dynamics.}$$

The saddle-node cascade changes the set of finite critical values $C_{J,i}(s)$; crossing those values can reorganize the accessible-region topology; and the periodic families of Section 6 are confined to the resulting Hill regions. The present computation demonstrates this connection through a resolution-checked representative transition rather than a complete topological classification.

8 Discussion

The results reveal a hierarchy of structures linking the geometry of the four-primary central configuration to the dynamics of the infinitesimal body. The central-configuration set determines which primary geometries are continuously accessible; along those components the restricted equilibria undergo folds and changes of stability; the resulting centre modes generate periodic-orbit families; and the associated critical Jacobi levels organize the energetically accessible regions. We discuss these implications in turn, together with the limits within which they have been established.

8.1 Central-configuration topology and equilibrium bifurcation structure

The principal structural result is that equilibrium multiplicity cannot be interpreted independently of the component structure of the underlying central-configuration set. On the fixed slice

$$(a, b) = (0.9, -0.1),$$

the two reference configurations with five and nine restricted equilibria belong to numerically distinct components of the regular zero set $\psi = 0$. Restricting these components to positive masses produces two different admissible arcs. Consequently, the difference between the five- and nine-equilibrium reference cases cannot be interpreted as the result of a single bifurcation sequence connecting the two sampled configurations.

This observation changes the role of the primary central configuration in the restricted problem. Rather than serving only as a fixed background geometry, its component structure determines which equilibrium branches can coexist along a continuous deformation and which sequences of bifurcations are accessible on that deformation. On the first admissible arc the interior multiplicity sequence is

$$5 \to 7 \to 5,$$

before the positive-mass problem approaches a mass-vanishing boundary. On the second arc the sequence is

$$7 \to 9 \to 11 \to 9 \to 7 \to 5.$$

Thus the equilibrium count is not a monotone measure of deformation or asymmetry; it depends on the ordering of individual folds along the admissible component.

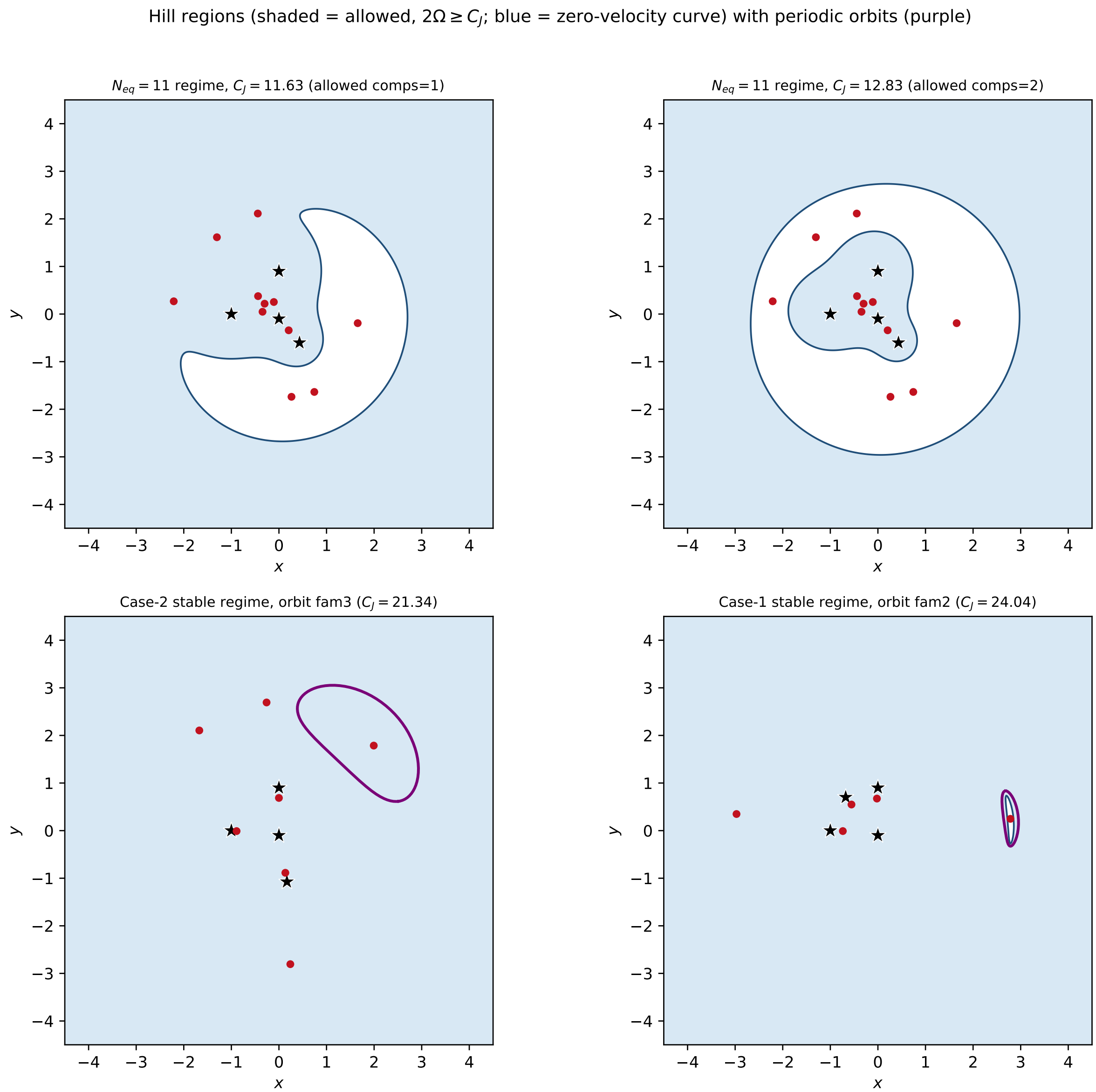

Figure 16: Hill regions (shaded: allowed, $2\Omega \geq C_J$; blue: zero-velocity curve). Stars denote primaries, red points equilibria, and purple curves periodic orbits. Top: the eleven-equilibrium regime at $C_J = 11.63$ (left) and $C_J = 12.83$ (right), bracketing a small cluster of finite critical Jacobi levels; the allowed set changes from one connected component to two across this interval as a neck closes. Bottom: representative Floquet-stable periodic orbits in the Case-2 (family 3, left) and Case-1 (family 2, right) regimes, each shown at its own Jacobi constant.

The eleven-equilibrium interval illustrates why the scalar count alone is an incomplete description. The pair B_a created at the $9 \to 11$ fold F_4 is not the pair destroyed at the subsequent $11 \to 9$ fold F_5. Instead, the pre-existing pair B_b is destroyed at F_5, while B_a persists until F_6. The two nine-equilibrium intervals on either side of the eleven-equilibrium window therefore have identical multiplicity but different branch populations. A return to a previous value of $N_{\rm eq}$ does not imply a return to the same equilibrium configuration.

The mechanisms responsible for changes of multiplicity must also be kept distinct. The seven interior transitions are bifurcations of the restricted equilibrium equations with all four primary masses positive. They satisfy the numerical degeneracy and nondegeneracy conditions for generic saddle-node folds and exhibit the expected square-root separation law. By contrast, the Case-1 $5 \to 4$ transition occurs as $\rho_2 \to 0$ near the boundary of the admissible primary set. It is therefore a parameter-boundary degeneration of the four-primary problem rather than an interior saddle-node bifurcation of the restricted dynamics.

These results suggest a broader organizing principle: in restricted problems with sufficiently general primary geometries, the component structure of the admissible primary-configuration set may organize the accessible restricted bifurcation sequences. The present calculation supports this interpretation on one two-dimensional slice only; whether it persists in the full four-parameter admissible set or in other non-symmetric restricted problems remains an open question.

8.2 Stability and nonlinear periodic dynamics

The equilibrium continuation also shows that spectral stability is not restricted to isolated symmetric configurations. Both admissible components contain finite intervals on which selected libration points are doubly elliptic. On the reliably tracked Case-1 branch this occurs over approximately

$$s \in [0.643, 0.835],$$

while the Case-2 component contains two distinct stable branches over overlapping parameter ranges. In particular, two different spectrally stable equilibria coexist near

$$s \simeq 0.196,$$

one close to a 5:2 centre-frequency commensurability and the other close to 2:1. This coexistence reinforces the conclusion that neither the arc coordinate s nor the equilibrium count by itself uniquely characterizes the local dynamics; equilibrium branch identity is also required.

The centre modes of the selected equilibria organize the local nonlinear periodic dynamics. At each parent used in Section 6, the selected imaginary eigenvalue satisfies the exact non-resonance condition required by the Lyapunov centre theorem. The ten computed families are therefore Lyapunov families in the local theorem sense, including those continued from saddle$\times$centre parents. What differs is their stability.

For the selected doubly-elliptic parents, the computed families retain their nontrivial Floquet multipliers on the unit circle throughout the continued intervals and are therefore Floquet stable in the sense defined in Section 6.3. This is a periodic-orbit statement and should not be interpreted as a proof of nonlinear or KAM stability. Conversely, families 5–6 originate from saddle$\times$centre parents. Their centre directions still generate valid Lyapunov families, but the continued periodic orbits possess a strongly real reciprocal Floquet pair and are correspondingly real unstable. The existence of a local periodic family and its Floquet stability are therefore logically distinct questions.

The low-order centre-frequency near-commensurabilities provide useful locations at which to examine possible changes in periodic-orbit stability, but they must not be confused with exact resonances. Families 7–8 are continued from the Case-1 parent near 3:2, while families 9–10 use the distinct Case-2 parent near 5:2. In both cases the parent frequency ratio differs from the exact rational value, so the Lyapunov-centre non-resonance condition remains satisfied.

The most pronounced nonlinear stability feature occurs on family 10. Its nontrivial Floquet pair approaches the period-doubling collision at

$$\nu = -2$$

to within approximately

$$|\nu + 2| \simeq 10^{-7}.$$

An ultra-fine amplitude sweep with step 8×10^{-4} resolves the minimum and shows the index turning back without crossing the threshold. The computed family therefore provides no evidence of a period-doubling bifurcation over the continuation interval investigated. Family 8 also approaches the same boundary, but only to within approximately 2×10^{-3}.

This negative result should be interpreted narrowly. The multiplier scans show no crossing of the selected low-order conditions

$$\nu = 2, \qquad \nu = 0, \qquad \nu = -1, \qquad \nu = -2$$

along the computed portions of the ten families. We have not performed an independent exhaustive search for all secondary periodic families, nor does the absence of a crossing on these continuation intervals exclude period-doubled or resonant solutions elsewhere in parameter or phase space.

The Jacobi and Hill-region calculations provide the geometric context for these periodic motions. The equilibrium branches determine a ladder of critical Jacobi values, and changes across these levels can reorganize the accessible region. The representative Case-2 calculation demonstrates a net change from one to two connected Hill components across a cluster of three nearby critical levels. The periodic examples then show selected Floquet-stable families embedded within the allowed regions at their own Jacobi constants. The Hill geometry determines where motion at a given energy is possible, but it does not determine the Floquet stability of the orbit contained within that region.

8.3 Relation to previous models, scope and limitations

The present problem differs structurally from the symmetric restricted five-body models most commonly treated in the literature. In those models, reflection, permutation, or rotational symmetry constrains the primary geometry to a prescribed low-dimensional family. For example, Ollongren's symmetric restricted five-body configuration supports nine libration points, including stable equilibria [8], while the ring problem and other symmetric formulations exploit rotational or reflection structure to organize their equilibrium and periodic-orbit families [5, 13, 14]. Symmetric four-body central configurations such as the Caledonian and kite families similarly restrict the primary configuration to a special geometric locus [11, 3].

The distinction in the present problem is not merely that the equilibrium count is larger. On the investigated non-symmetric slice, the admissible central-configuration set itself separates into different components, and the restricted equilibrium multiplicity must therefore be interpreted component by component. The eleven-equilibrium state exceeds the nine-equilibrium counts reported in the symmetric restricted five-body examples cited above, but its more significant feature is that it arises within a continuously resolved saddle-node cascade and contains a branch exchange. Likewise, the periodic families are continued from asymmetrically located equilibria using a general shooting formulation and full-monodromy Floquet analysis rather than being constructed through an assumed reflection symmetry. The comparison with previous work is therefore primarily structural rather than a quantitative comparison at matched masses.

Several limitations delimit these conclusions. First, all primary continuation is carried out on the fixed slice

$$(a, b) = (0.9, -0.1).$$

The numerically distinct components identified there need not remain disconnected when a and b are allowed to vary. Determining the topology of the complete admissible set would require continuation in the full four-dimensional geometric parameter space.

Second, equilibrium completeness is numerical rather than analytic. The reported multiplicities are supported by deterministic grid searches, fixed-seed random starts, local searches around the primaries, strict residual tests, and an expanded re-search of the eleven-equilibrium configuration. These procedures provide strong numerical evidence for the reported counts under the stated search protocol, but they do not provide an analytic upper bound on the number of real equilibria.

Third, branch tracking becomes unreliable near the mass-vanishing limits where some equilibria recede rapidly. Such points are explicitly flagged and excluded from branch-dependent stability, centre-frequency, Jacobi, and periodic-orbit claims. The reliably tracked intervals are therefore shorter than the full ranges over which individual numerical roots may still be found.

Fourth, the periodic-orbit calculations use natural amplitude continuation rather than pseudo-arclength continuation. The reported 198 solutions therefore represent the portions of ten families accessible while the chosen amplitude remains a useful continuation coordinate; they are not claimed to be maximal families. Likewise, no systematic continuation of secondary resonant or period-doubled branches has been undertaken.

Fifth, the Hill-region calculation is representative rather than exhaustive. The connectivity change reported in Section 7 is resolution-checked on 300^2, 400^2, and 500^2 grids, but the selected Jacobi interval spans three nearby equilibrium critical values. The calculation therefore establishes a net $1 \to 2$ connectivity change across that part of the critical-level ladder rather than assigning the change to one particular critical equilibrium. A complete topological classification would require resolving the Hill geometry on both sides of every relevant critical level.

Finally, the stability results have deliberately limited meaning. We establish spectral stability of equilibria and Floquet stability of periodic orbits. We do not establish nonlinear, orbital, or KAM stability, and finite-time numerical boundedness is not used as a substitute for any of these notions.

Within these limits, the principal structural conclusions are robust: on the investigated slice the admissible central-configuration component structure organizes the accessible equilibrium bifurcation sequences; those sequences contain generic folds, stable-equilibrium intervals, and a branch-exchange eleven-equilibrium regime; and selected centre modes extend to well-resolved Lyapunov periodic families whose Floquet behaviour and Hill-region setting can be followed quantitatively.

9 Conclusions and outlook

We have studied how the equilibrium structure, linear stability, and local periodic dynamics of an infinitesimal fifth body change as a general non-symmetric four-primary central configuration is continuously deformed. Rather than comparing isolated primary configurations, the analysis follows the admissible central-configuration set itself and resolves the restricted equilibria, their bifurcations, and selected periodic-orbit families along that set.

Three conclusions emerge.

First, on the investigated slice

$$(a, b) = (0.9, -0.1),$$

the admissible central-configuration set is numerically separated into two distinct components containing the Case-1 and Case-2 reference configurations. The five- and nine-equilibrium cases therefore do not lie on a single fixed-(a, b) continuation family. This establishes that equilibrium multiplicity must be interpreted together with the component structure of the underlying primary configuration: different sampled counts need not be connected by a bifurcation sequence on the same admissible component. The result is specific to the investigated slice and does not establish disconnection in the full four-parameter admissible set.

Second, the restricted equilibrium structure along these components is organized by seven generic saddle-node folds. The corresponding interior multiplicity sequences are

$$5 \to 7 \to 5$$

on the Case-1 component and

$$7 \to 9 \to 11 \to 9 \to 7 \to 5$$

on the Case-2 component. The Case-1 arc additionally exhibits a mass-vanishing boundary degeneration as $\rho_2 \to 0$, which is distinct from the interior folds. The eleven-equilibrium interval is structurally significant because it contains a branch exchange: the pair B_a created at the $9 \to 11$ fold is not the pair B_b destroyed at the following $11 \to 9$ fold. Thus two configurations can have the same equilibrium count while containing different sets of equilibrium branches. The scalar multiplicity is therefore only a coarse description of the underlying equilibrium structure.

Third, this equilibrium organization is inherited by the local dynamics. Both admissible components contain reliably resolved intervals of spectrally stable equilibria, and the Case-2 component supports two distinct spectrally stable equilibrium branches over overlapping parameter ranges. From selected simple centre modes we computed ten Lyapunov periodic-orbit families comprising 198 corrected solutions. The families continued from the selected doubly-elliptic parents remain Floquet stable over the computed intervals, whereas the families continued from selected saddle×centre parents possess a real reciprocal Floquet pair and are real unstable.

Low-order centre-frequency ratios encountered along the stable branches are near-commensurabilities rather than exact resonances at the parent equilibria. Families 7–8 sample the Case-1 region near 3:2, while families 9–10 sample the distinct Case-2 region near 5:2. Family 10 approaches the period-doubling boundary extremely closely,

$$\min |\nu + 2| \simeq 10^{-7},$$

but an ultra-fine amplitude sweep resolves a turn-back rather than a crossing. No period-doubling bifurcation is therefore detected on the computed portion of that family. More generally, no crossing of the selected low-order multiplier conditions is found along the ten continued families; this does not exclude secondary periodic solutions elsewhere in parameter or phase space.

The equilibrium structure also determines a hierarchy of critical Jacobi levels. A representative calculation in the eleven-equilibrium regime shows a resolution-robust net change in Hill-region connectivity from one component to two across a small cluster of three nearby equilibrium critical levels. Representative Floquet-stable periodic orbits remain inside the corresponding allowed regions at their own Jacobi constants. This provides a concrete geometric link between the equilibrium bifurcations, the critical energy levels, the accessible Hill geometry, and the periodic dynamics without implying that every critical level produces a change in component count.

Taken together, the results support a broader interpretation: in a non-symmetric restricted problem, the topology of the admissible primary-configuration set can itself act as an organizing structure for the restricted dynamics. On the present slice it determines which equilibrium branches can coexist and which bifurcation sequences are accessible; those branches in turn determine the available centre modes, critical Jacobi levels, and local periodic families. Whether this organizing role persists in the full parameter space is a natural question for further study.

Several extensions follow directly from the present analysis. The most important is continuation in the complete four-parameter admissible central-configuration set, which would determine whether the two components identified here remain disconnected when a and b are also allowed to vary. A second direction is pseudo-arclength continuation of the periodic-orbit families themselves, together with targeted continuation of secondary branches near the observed low-order commensurabilities and the near-period-doubling point. A third is a systematic Hill-region classification across individual critical Jacobi levels rather than the representative connectivity calculation performed here. Finally, comparison with symmetric restricted five-body models at matched or otherwise controlled mass configurations would help separate effects caused by mass distribution from those caused specifically by the geometry and topology of the admissible central-configuration set.

Data and code availability

The model, continuation, bifurcation, stability, and periodic-orbit computations, together with the scripts that generate every table and figure in this paper deterministically from the verified result files, are organized as a single release archive. Every numerical result was produced with *Python 3.14.5* and *NumPy 2.5.0*, *SciPy 1.17.1*, *Matplotlib 3.10.9*, and *pandas 3.0.3*; integrations use `scipy.integrate.solve_ivp` with the `DOP853` method at $\mathrm{rtol} = \mathrm{atol} = 10^{-12}$, and all random seeds are fixed, so the pipeline is designed to be deterministically reproducible within the pinned software environment. The complete code and verified result files are available in a public repository, with an immutable snapshot archived for the published version. The identifiers are: *repository:* `https://github.com/drmshoaib/4-1-body-problem`; *release tag:* `v1.0-paper-submission`; *archive DOI:* [to be added]. Full numerical settings and per-file provenance are documented in the repository README.

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